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Mining association rules and sequential patterns from electronic prescription databases

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Abstract

Over the years many scientific studies have been published in Medicine to evaluate, understand and predict the effects of introducing new medications. However, those studies draw conclusions from small samples, due to the difficulty and cost of retrieving large quantities of data through questionnaires. Thanks to the growing trend in prescription process automation, large amounts of medical data are now stored in databases that can later be explored to discover potentially useful information. Electronic Prescription data can be analyzed to improve prevention, diagnosis and treatment of diseases, optimize resources, and promote patient safety. This dissertation presents a methodology to discover association rules and frequent sequences in databases of electronic prescriptions using the Apriori and PrefixSpan algorithms. The methodology was used to characterise the Portuguese population prescribed with anticoagulants. This study enabled (a) an assessment of the adoption of novel oral anticoagulants, including the identification of predictive factors associated with discontinuation or changes of prescribed medication, (b) discovery of causal association rules between medications, and (c) characterization of frequent patterns associated with the consumption of anticoagulants. The main conclusion of this work is that data mining techniques can be applied to electronic prescription databases to extract knowledge which can latter support decision-making in public health.

Keywords

Anticoagulant prescriptions analysis; Electronic prescriptions mining; Frequent and sequential patterns; Data mining in health applications

Resumo

Ao longo dos anos têm sido publicados estudos científicos em Medicina para avaliar, compreender e prever os efeitos da introdução de novos medicamentos. Contudo, esses estudos retiram conclusões a partir de pequenas amostras, devido à dificuldade e ao custo de recolher grandes quantidades de dados através de questionários. Graças à tendência de automatização dos processos de prescrição, vastos conjuntos de dados médicos começaram a ser armazenados em bases de dados que podem posteriormente ser exploradas e analisadas de modo a descobrir informação escondida e potencialmente útil. Os dados de prescrições eletrónicas podem ser analisados para melhorar a prevenção, diagnóstico e tratamento de doenças; otimizar recursos; e promover a segurança dos pacientes. Esta dissertação apresenta uma metodologia para descobrir regras de associação e sequências frequentes em bases de dados de prescrições eletrónicas, usando os algoritmos Apriori e PrefixSpan, bem como a sua aplicação à análise da população portuguesa prescrita com anticoagulantes. Apresenta-se a metodologia à população portuguesa prescrita com anticoagulantes. O estudo realizado permitiu (a) avaliar a adoção de novos anticoagulantes orais, incluindo a identificação de fatores preditivos associados com descontinuação ou mudanças na medicação, (b) descobrir regras de associações causais entre medicações, e (c) caracterizar padrões frequentes associados com o consumo de anticoagulantes. A conclusão principal deste trabalho é que a aplicação de técnicas de prospecção de dados a bases de dados de prescrições médicas permite extrair conhecimento que pode posteriormente ser usado para apoiar tomadas de decisão em saúde pública.

Palavras Chave

Análise de prescrições de anticoagulantes; Análise de prescrições de medicamentos; Padrões Frequentes e sequenciais; Prospecção de dados de saúde

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Acronyms

AR	Association Rule
ARM	Association Rule Mining
CEP	Clinical Event Package
CP	Clinical Pathway
CPE	Care Pathway Explorer
CSP	Concurrent Sequential Patterns
CRISP-DM	Cross-Industry Standard Process for Data Mining
DDD	Defined Daily Dosage
DOAC	Direct Oral Anticoagulant
DM	Data Mining
EMR	Electronic Medical Records
FS	Frequent Sequences
GPD	Guided Process Discovery
IM	Interesting Measure
KDD	Knowledge Discovery in Databases
LPM	Local Process Model
MFI	Maximal Frequent Itemset
MP	Minimal Pattern
MINSUP	Minimum Support
NOAC	Novel Oral Anticoagulant
PIR	Potential Interesting Rules
SDCE	Same Day Concurrent Event

SEMMA Sample Explore Modify Model Assess

SM Subjective Measures

SP Single Pattern

SPM Sequential Pattern Mining

SRP Structural Relation Patterns

1. Introduction

In 2010 the Portuguese government deployed an electronic platform to support the prescription, dispense and billing of medications, with the objective of not only making the system more efficient and secure, but also to promote better quality and rationality in prescription and dispense of medications[1]. This strategy meant that large pools of data started being collected and stored in databases, which can subsequently be queried and analyzed in order to unravel hidden and potentially useful information.

These large volumes of data meant that traditional statistical methods were no longer appropriate to analyze the data. Thanks to advances in both computer technology and artificial intelligence, new automated techniques to extract knowledge from large sets were developed, in particular Data Mining (DM) techniques which enabled the extraction of knowledge from analyzed products, that can assist in decision making and can deal with noisy or missing data [Rogalewicz and Sika, 2016]. DM is a subdiscipline of computer science focused on finding hidden relations and on giving summaries, through patterns and models, from large pools of data in a way that is understandable and useful [Hand et al., 2001]. One of the first examples of DM applied in real-world scenarios relates to market basket analysis. These techniques enabled retailers to optimize product placement and promotions by uncovering associations between items, through the identification of item-sets that occur frequently together in transactions. While data mining techniques have been applied successfully in other types of databases [Ngai et al., 2011, Rygielski et al., 2002, Antonie et al., 2001, Romero and Ventura, 2007], their application over electronic prescription databases has not been significantly explored. However, we believe that data mining algorithms can be applied to prescription databases to uncover interesting patterns (e.g., co-occurrences between different medications, or sequences of medications appearing frequently and corresponding to common treatment regimes). In particular, perhaps these techniques can be used to help explain the adoption rates of the newer generation of anticoagulants in Portugal, using the prescription data stored in the national healthcare system database, in particular by identifying associations and patterns latent in the prescription records.

Anticoagulants are an interesting set of medications to study given not only their present significant impact in term of costs to the Portuguese national healthcare system, but also due to population aging and the fact that people are being increasingly prescribed [2] with anticoagulants. They are the 3rd pharmacotherapeutic group with the highest weight in the drug expenditure (7.7 % in 2017) of the Portuguese

[1] http:/www.infarmed.pt/index.html

[2] http://www.infarmed.pt/web/infarmed/entidades/medicamentos-uso-humano/monitorizacao-mercado/relatorios/ambulatorio

national health service [Alves da Costa et al., 2017], appearing also frequently in the corresponding medical prescriptions database (i.e., it will be interesting to look at other medications that appear frequently together with anticoagulants for the same patients, and to look at Frequent Sequences (FS) of prescriptions involving anticoagulants). Traditional anticoagulants (e.g., Warfarin, other Coumarins, and Heparins) are in widespread use but, since the 2000s, a number of new agents have been introduced that are collectively referred to as Novel Oral Anticoagulant (NOAC) or Direct Oral Anticoagulant (DOAC). These agents include direct thrombin inhibitors (e.g., Dabigatran Etexilate) and factor Xa inhibitors (e.g., Rivaroxaban or Apixaban) and they have been shown to be as good or possibly better than the traditional anticoagulants, at the same time with less serious side effects. The newer anticoagulants (NOAC / DOAC) are nonetheless more expensive than the traditional ones, and they should be used with care in patients with kidney problems.

In the context of my M.Sc. research project, I started with a statistical characterization of a dataset containing electronic prescriptions from the Portuguese national healthcare system comprising prescription data between the years 2013 and 2016, to have a global view on the anticoagulants prescription situation. I then further explored the dataset using association rules (i.e., there is a list of antecedent medications and a list of consequent medications), causal rules (i.e., similar to association rules but the association is stronger between the antecedent and consequent, namely the fact that taking the antecedent implies that the consequent is also taken) and frequent sequences between prescribed medications.

1.1 Objectives

The main objectives of this work can be summarized as follows:

- Development of a methodology to prepare data for mining algorithms.
- Perform a characterization of the Portuguese population prescribed with anticoagulants, and assess the adoption of novel oral anticoagulants, including predictive factors associated with discontinuation or changes.
- Discover association rules, including causal associations, between medications from electronic prescription databases, and discover sequential patterns in medication prescriptions.

1.2 Methods

Most academic studies in the DM field can be divided in two main paradigms depending on its the goals: Verification and Discovery. In Verification the system is limited to verifying the user hypothesis, while in Discovery the system autonomously finds new patterns. Discovery can be further divided,

depending on whether the data is known and labeled (Supervised Learning), and in this case we have classification and regression models, or if the inputs and outputs are unknown (Unsupervised Learning), where we can use clustering and Association Rule Mining (ARM) [Ravindra Babu et al., 2013, Fayyad et al., 1996a].

ARM consists on the discovery of implications in the form of $A \rightarrow B$, where A is the antecedent and B the consequent are disjoint item-sets, on an analyzed data set. Following the objectives described in Section 1.1, my main approach falls into the category of the Association Rule Mining. To guide my work in the this research, I propose the use of the Knowledge Discovery in Databases (KDD) process to unearth non trivial regularities, relationships and schemes from the data [Fayyad et al., 1996b]. Other methodologies such as Cross-Industry Standard Process for Data Mining (CRISP-DM) and Sample Explore Modify Model Assess (SEMMA) were also explored, but the three, at their core, can be mapped. We mainly choose KDD due to the fact that CRISP-DM only adds Business Understanding and Deployment phases [Azevedo and Filipe Santos, 2008], which are not applicable to this research, and SEMMA is poorly supported with documentation and implementation guides [NIAKŠU, 2015].

According to [Fayyad et al., 1996a], the KDD process consists of *using the database along with any required selection, pre-processing, sub sampling, and transformations of it; to apply data mining methods (algorithms) to enumerate patterns from it; and to evaluate the products of data mining to identify the subset of the enumerated patterns deemed "knowledge"*. By being an interactive and iterative process it can involve significant iteration and may contain loops between any of its five phases:

1. **Selection**: This phase consists on creating a target data-set, or focusing on a subset of variables or data samples. In this study, we focused on the prescriptions from Portuguese patients that had at least one prescription of anticoagulants between the years of 2015 and 2016.

2. **Pre-processing**: This phase consists on the target data cleaning and pre-processing in order to obtain consistent data. Due to the longitudinal nature of this study, special care was taken to carefully examine and consolidate the attribute fields.

3. **Transformation**: This phase consists on the transformation of the data using dimensionality reduction or transformation methods. We used external information about medication treatments, in this case the concept of Defined Daily Dosage (DDD), to transform a database of electronic prescriptions into one that reconstructs the time intervals for which each patient was subjected to a certain treatment.

4. **Data Mining**: This phase consists on the searching for patterns of interest in a particular representational form, depending on the DM objective. The algorithms Apriori and PrefixSpan were applied to discover association rules and frequent sequences between medications.

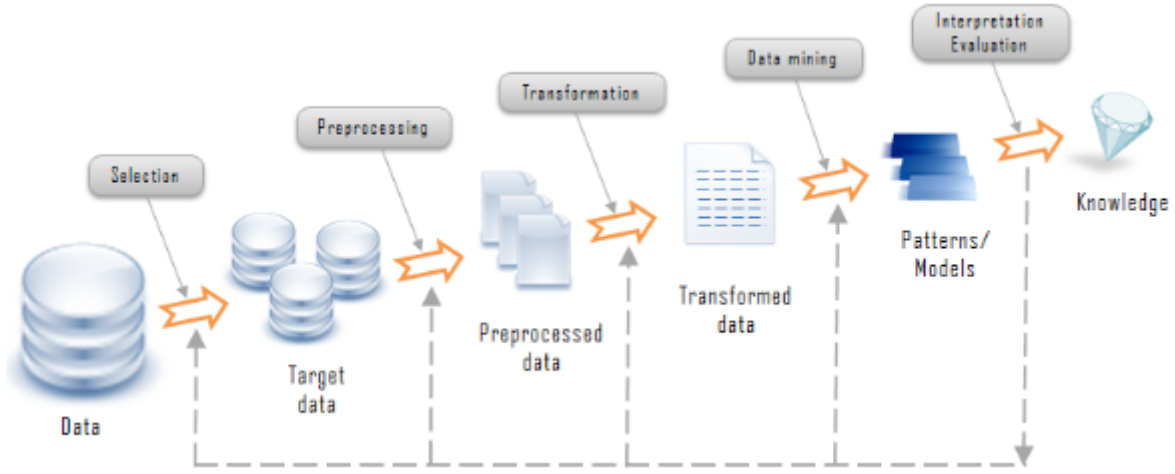


Figure 1.1: Process for knowledge discovery from a database.

5. **Interpretation**: This phase consists on the interpretation and evaluation of the mined patterns. Based on the framework Rule Changing + Relevance Feedback, we used evaluation measures Support, Confidence and Lift, combined with concepts of Unexpectedness, Novelty and Cohort Studies, to evaluate and rank the results.

The strength of the KDD process relies in its explicit simplicity, which makes it applicable to almost all knowledge discovery domains. However, only a generic guideline is provided, with no formal methodology or accompanying tool-set. Nevertheless, this process model is one of the most referenced and used for general Knowledge Database Discovery purposes, and it became the base model for other more detailed models like CRISP-DM and SEMMA [NIAKŠU, 2015].

1.3 Contributions

In this thesis we obtained the following results:

1. Definition of a methodology to transform a set of medical records into a suitable database for data mining tasks, envisioning the discovery and exploration of relationships between prescribed medications.

2. From the application of the Apriori algorithm, it was possible to observe that the rules associated with the male patients have a greater lift when compared with the female ones. Also, it was interesting to note that not only the female top rules contain 33% more medications than the male's, but also that the NOAC Rivaroxaban already appears in the top rules. The comparison between the Association Rule (AR) from Lisbon and Porto, the two biggest districts in terms of population, showed that Porto produces AR with a much larger lift.

3. From the application of the PrefixSpan algorithm, it was possible to observe that in age group 0-44 Warfarin appears linked, as expected, to medications used to treat hypertension and cholesterol, but also to less expected treatments like arrhythmia. Even more surprising was the connection found, in age group 65-74, between NOAC Rivaroxaban and medications used in insomnia.

1.4 Document Structure

This dissertation is divided in six chapters. In Chapter 1 the problem targeted for study in the context of this M.Sc. thesis was introduced, together with the main paradigms and the research methodology that was adopted. Chapter 2 introduces the concepts of AR and FS, and describes the main algorithms to find them, including the associated evaluation methods. Chapter 3 presents advanced approaches to the problem of mining sequential data, including previous work done in health record databases. Chapter 4 provides a detailed step-by-step explanation on the methods used for this work, including the pre-processing, data mining and evaluation modules. Chapter 5 makes an initial characterization of the data, and then depicts the results obtained with the proposed methodology, using adequate visualization techniques. Finally, Chapter 6 provides a reflection on the whole work that was developed, including problems encountered and suggestions for future work.

2. Fundamental Concepts

This chapter introduces the concepts and associated algorithms of ARM in Section 2.1, and Sequential Pattern Mining (SPM) in Section 2.2 to this work. Then, Section 2.3 provides an overview on the different metrics, methods and evaluation frameworks available to identify Potential Interesting Rules (PIR). Finally, Section 2.4 provides a brief summary about the introduced concepts.

2.1 Association Rule Mining

Association Rule Mining (ARM) is a data mining technique concerned with finding all large itemsets (i.e., collections of items) that satisfy both syntactic and support constraints [Agrawal et al., 1993]. Association rules encode strong co-occurrence patterns, although these do not imply causality between items. To define association rules, let us assume $I = \{i_1, i_2, \ldots, i_n\}$ to be a set of n attributes called items, and T to be a set of transactions called a database. With these definitions, we can say that an association rule is an implication:

$$X \Rightarrow I_j \tag{2.1}$$

In the previous expression, X is a subset of I and I_j is a subset $I_j \subseteq I$ that is absent in X. Restrictions on the items that appear in Expression 2.1 are syntactic constraints, while support restrictions are usually expressed by the evaluation metrics of support, confidence and lift (see Section 2.3.1.1).

A transaction t can be represented by a unique identifier and a binary vector that represents the occurrence/absence of a subset of items in I. A transaction t satisfies X if $\forall I_k \in X$, $t[k] = 1$. The association rule in Expression 2.1 is satisfied in T with confidence level $c \in [0, 1]$ iff at least a percentage c of transactions in T that contain I_j (rule's consequent) also contain contain X (rule's antecedent). If association rules are able to satisfy user-defined requirements such as minimum confidence (min-confidence) and minimum support (min-support) of the antecedent, then they are called strong rules.

A naïve approach to mine association rules would involve generating all possible rules and then calculating the support and confidence for each one, pruning the rules that fail to meet the min-support and min-confidence thresholds. However, this would be impractical since the total number of possible rules grows exponentially with the number of items n in the database [Tan et al., 2005], according to:

$$O(n) = 3^n - 2^{n+1} + 1 \tag{2.2}$$

To improve performance, [Agrawal and Srikant, 1994] proposed that the association rule mining problem could be divided in two steps:

1. **Mining frequent item-sets**: Generate all item-sets that have fractional transaction support above min-support. This step is the most computationally expensive with time complexity $\mathcal{O}(N \times M \times w)$, where N is the number of transactions, where $M = 2^k - 1$, with k as the number of items, is the number of the candidate item-sets, and where w is the maximum transaction width [Tan et al., 2005].

2. **Rule generation**: The objective is to extract high confidence rules from the frequent item-sets found in the previous step. To extract these rules the algorithm iterates through the large item-sets l (i.e., item-sets with min-support) and for each, finds all the non-empty subsets of l. Each of these subsets a output a rule of the form $a \Rightarrow (l - a)$ if the ratio of support(l) to support(a) is at least min-confidence. Thus, each generated rule is a binary partition of a frequent item-set.

Despite this division, mining association rules still remains computationally expensive. Next we present two algorithms that explore this division to improve efficiency, namely Apriori and FP-Growth.

2.1.1 The Apriori Algorithm

[Agrawal and Srikant, 1994] proposed an item-set generation strategy called Apriori to address the complexity problem, enabling a reduction of candidate item-sets before counting their support values. Apriori is an interactive breadth-first search algorithm using a generate-and-test strategy to mine frequent item-sets for Boolean association rules. It is based on the principle that *if an item-set is frequent, then all of its subsets must also be frequent* [Tan et al., 2005]. Leveraging on this principle, Apriori can prune candidate item-sets with infrequent subsets without having to count their support. Figure 2.1 shows the reduction of the search space using Apriori principle. This holds true due to the anti-monotonic property of support:

$$\forall X, Y : (X \subseteq Y) \Rightarrow \mathrm{s}(Y) \leq \mathrm{s}(X) \tag{2.3}$$

In the previous expression, X and Y represent item-sets and $\mathrm{s}(X)$ represents the support associated with item-set X. Expression 2.3 denotes that the support of an item-set never exceeds the support of its subsets. Initially, the algorithm starts by determining the support of each item, thus finding the set of all frequent 1-item-sets. Next, the algorithm will iteratively generate new candidate k-item-sets using frequent (k-1)-item-sets found in the previous iteration. After counting the support of the newly generated candidate item-sets, the algorithm eliminates all those who fail to meet the minimum support

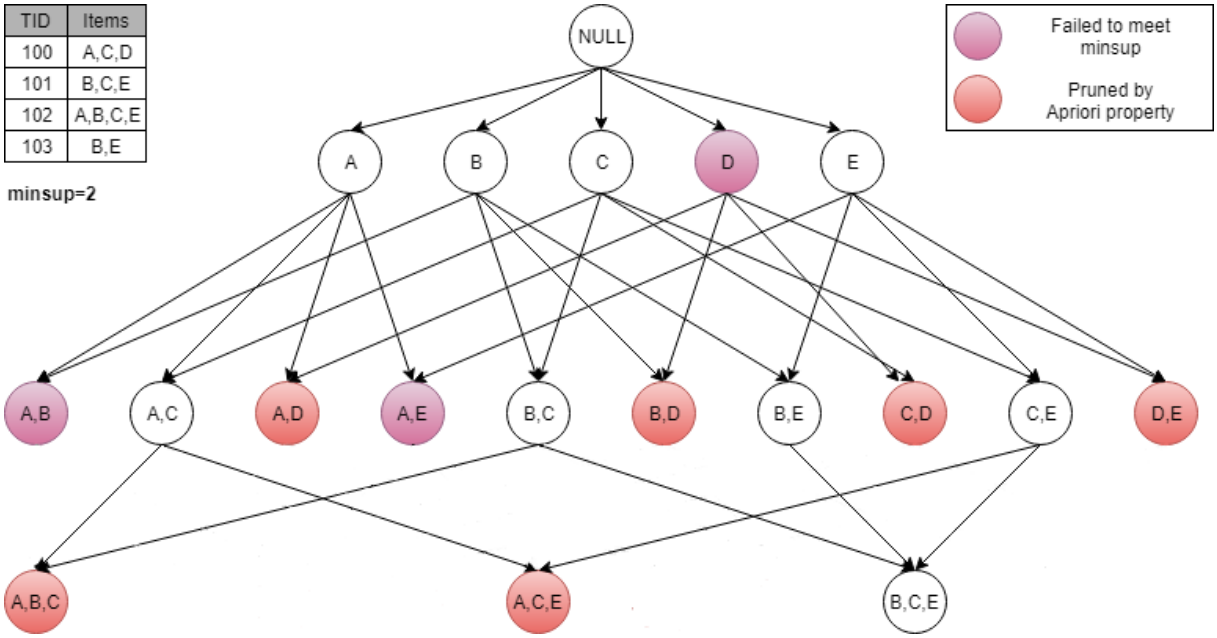


Figure 2.1: The Apriori principle for pruning candidate item-sets.

threshold. The remaining ones constitute the frequent item-set. The algorithm stops when there are no new frequent item-sets being generated through this procedure.

The advantages of Apriori are related to its simplicity and to the reduction of candidate item-sets, by applying the Apriori principle. However, a bottleneck still exists since multiple passes over the database are necessary. Moreover, the generation step can produce a very large number of candidate sets (i.e., lengthy patterns). Many extensions have nonetheless been proposed. For instance, AprioriTid [Agrawal and Srikant, 1994] is a variant that also uses the candidate generation step of the regular Apriori algorithm to determine its item-sets. However, the database is not used to count support after the first pass, and instead the set of candidate item-sets is used for this purpose. Compared to Apriori, this variant has better performance in the later passes. Another method called Apriori Hybrid [Agrawal and Srikant, 1994] combines the best of both proposals, using standard Apriori for initial passes and AprioriTid for the later ones.

2.1.2 The FP-Growth Algorithm

FP-Growth is a depth-first search algorithm that, unlike Apriori, does not use a generate-and-test approach. It was first proposed by [Han et al., 2000] as an efficient method for mining the complete set of frequent patterns by pattern fragment growth, using a highly condensed representation of the data called FP-tree. An FP-Tree is a data structure composed of nodes representing items, including a counter, and paths denoting a transaction. The FP-Growth algorithm is based on two steps:

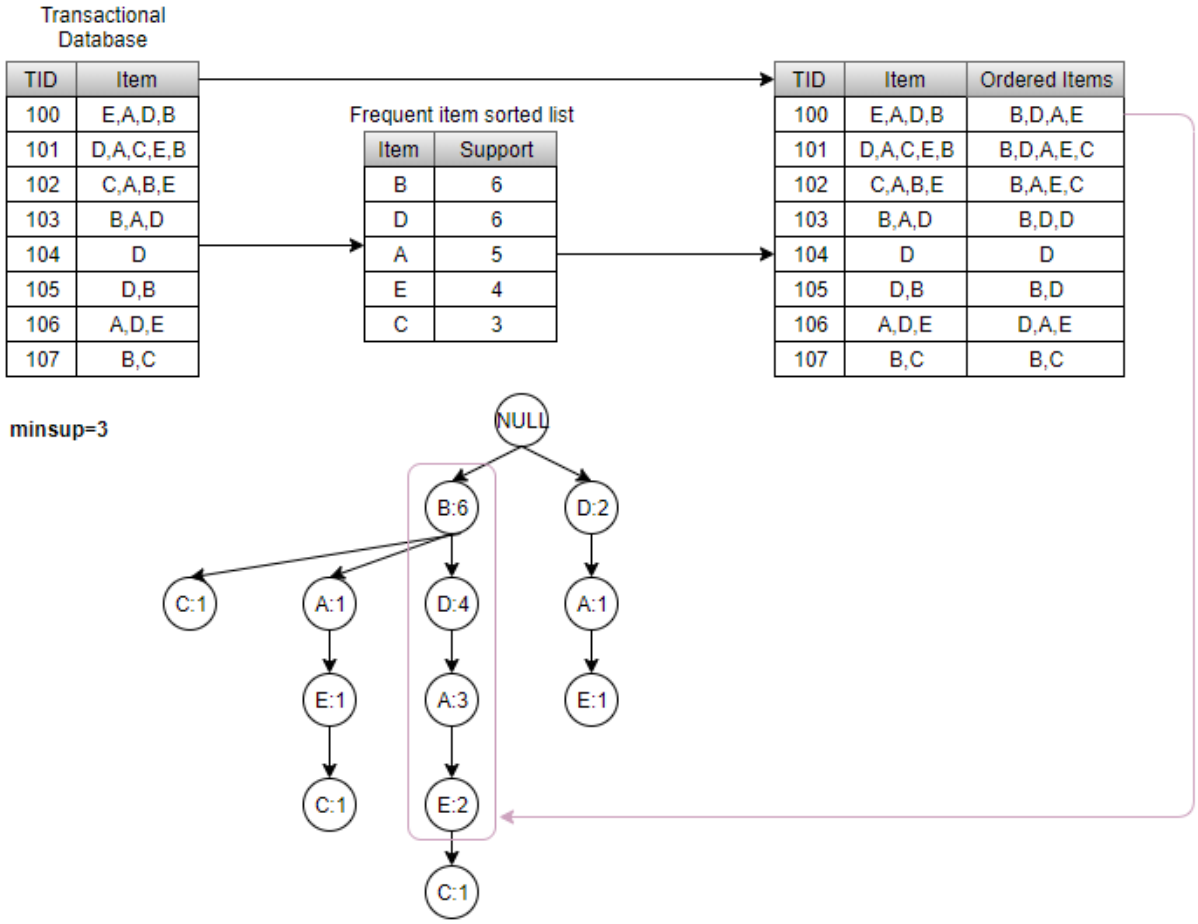


Figure 2.2: The FP-Growth algorithm.

1. **Building the FP-Tree**: First, one scan over the data is used to create a support-descending ordered list of frequent item-sets. Then, the tree construction starts by reading each transaction, in the order of the sorted list, and mapping it into a path in the FP-Tree (see Fig. 2.2). In this step only 2 scans over the database are made (one for counting the support of each item and possibly pruning the result, and another pass to build the tree in decreasing item support).

2. **Extract Frequent Item-sets**: The tree is now traversed to find all item-sets for each item. To do this, we need to find a conditional base pattern (CBP) for each pattern (i.e., prefix-paths in the FP-Tree which consist of a suffix pattern). From the CBP, a conditional tree is generated, which is recursively mined in the algorithm. If the size of the FP-tree is small enough to fit into main memory, then we can extract frequent item-sets directly from the structure in memory instead of making repeated passes over the FP-Tree stored on disk.

FP-Growth is faster than Apriori [S.Mythili and Shanavas, 2013], conserves complete Apriori information as well for frequent pattern mining, constructs a highly compact FP-Tree (i.e., overlapping paths) thus

reducing the cost of scans in subsequent mining methods, avoids the costly candidate generation step, and uses a divide-and-conquer method [Tan et al., 2005] to reduce the size of subsequent conditional pattern bases and conditional FP-Trees. However, frequent patterns that do not fit in memory impact performance significantly, similar to Apriori. The method is also not ideal for interactive mining systems (i.e., when changing the support threshold according to rules), nor is it suitable for incremental mining (i.e., avoid redoing mining on the whole database when an update occurs).

2.2 Sequential Pattern Mining

The SPM problem was first introduced by [Agrawal and Srikant, 1995] with basis on the following definition: *given a set of sequences, where each sequence consists of a list of elements and each element consists of a set of items, and given a user-specified min-support threshold, SPM concerns finding all of the frequent sub sequences, i.e., the sub sequences whose occurrences frequency in the set of sequences is no less than min-support*. Both ARM and SPM represent both intra and inter-transaction relationships between transactions. Although Apriori can be used for SPM [Patil and Patil, 2013], the complexity associated with the generation of rules lead to the appearance of new algorithms. Specifically, [Pei et al., 2001] proposed the Prefix-Projected Sequential Pattern Mining method, also known as PrefixSpan.

2.2.1 The PrefixSpan Algorithm

In brief, PrefixSpan is an efficient algorithm for mining sequential patterns in large databases with time-related knowledge. It is an example of a pattern growth algorithm, that examines only the prefix sub-sequences and projects only their corresponding postfix sub-sequences into a new database (projected database). In each projected database, sequential patterns are grown by exploring only frequent patterns. Thus the major cost in PrefixSpan is to build projected databases. Since items within a sequence can be listed in any order, without the loss of generality, we assume they are listed in alphabetic order, hence the sequence is unique. Before diving into the algorithm we need to define three concepts:

- **Prefix**: Given a sequence $\alpha = \langle e_1 e_2 ... e_n \rangle$, a sequence $\beta = \langle e'_1 e'_2 ... e'_m \rangle$ where $(m \leq n)$, is called a prefix of α iif: $e'_i = e_i$ for $(i \leq m-1)$, $e'_m \subseteq e_m$ and all items in $(e_m - e'_m)$ are alphabetical after those in e'_m.

- **Projection**: Given sequences α and β such that β is a sub sequence of α, i.e. $\beta \sqsubseteq \alpha$, a sub sequence α' of α is called a projection of α with respect to prefix β iif α' has prefix β and there exists no proper super-sequence α'' of α' (i.e., $\alpha' \sqsubseteq \alpha''$ but $\alpha' \neq \alpha''$) such that α'' is a subsequence of α and also has prefix β.

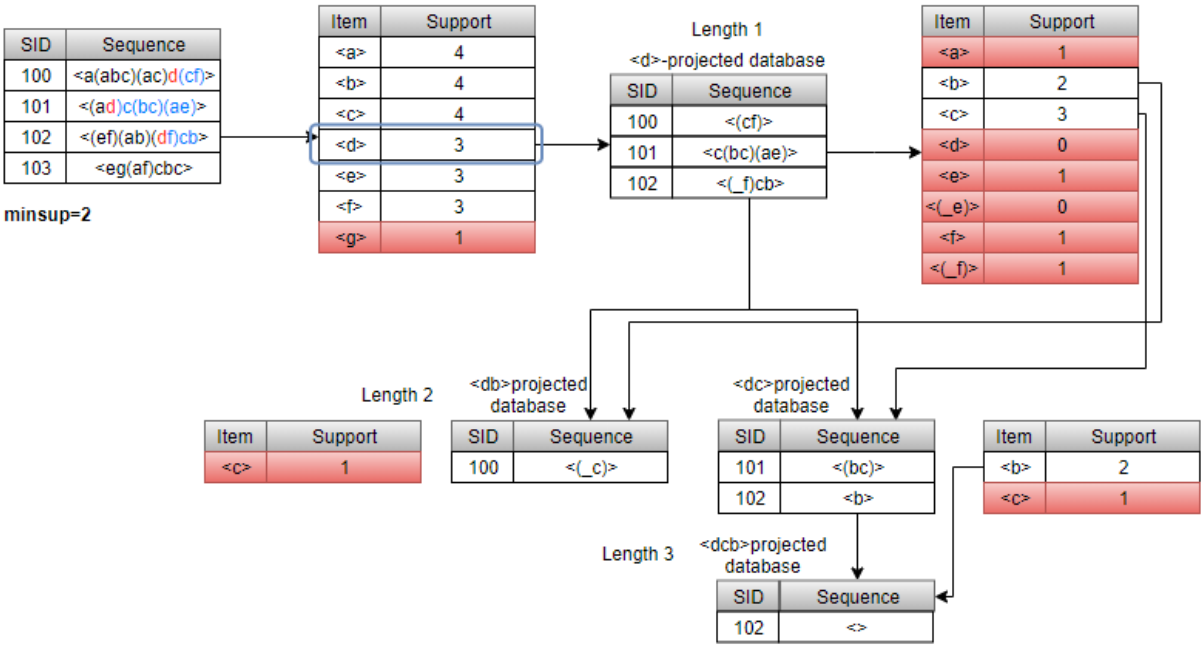


Figure 2.3: The PrefixSpan algorithm.

- **Postfix**: Let $\alpha' = \langle e_1 e_2 ... e_n \rangle$ be the projection of α with respect to prefix $\beta = \langle e_1 e_2 ... e_{m-1} e'_m \rangle$ where $(m \leq n)$. Sequence $\gamma = \langle e''_m e_{m+1} ... e_n \rangle$ is called the postfix of α with regards to prefix β, denoted as $\gamma = \alpha/\beta$ and where $e''_m = (e_m - e'_m)$. If β is not a subsequence of α, both the projection and the postfix of α with regards to β are empty.

The PrefixSpan algorithm receives as input a set of sequences S and the min-support threshold. Let α be a sequential pattern, L the length of α, and $S|\alpha$ the α-projected database if $\alpha \neq \langle\rangle$ and S otherwise. The algorithm executes in the following three steps:

1. Scan $S|\alpha$ once to find frequent item-set b so that b can be assembled to the last element of α to create a sequential pattern, or $\langle b \rangle$ can be appended to α to create a sequential pattern.

2. Append each frequent item in b to α, in order to form a sequential pattern α' that is then produced as output.

3. For each α' construct a α-projected database $S|\alpha'$ and return to Step 1.

PrefixSpan explores prefix-projection in SPM, enabling us to mine the complete set of patterns without having to generate candidate sequences. Also, since projected-databases keep shrinking, the process is more efficient than Apriori (see Fig. 2.3). The major cost lies in the construction of projected databases. An alternative to improve this step involves Bi-Level projections [Pei et al., 2001], where instead of projecting databases at every level, it only projects every two levels. This results in better performance when database is large and with low support threshold. If the database can be stored in memory,

then another efficient alternative for this step are pseudo projections, leveraging on pointers to refer to sequences in the database as a pseudo projection, instead of constructing it [Pei et al., 2001].

2.3 Evaluation Methods

From the vast amount of rules generated by DM techniques, only a small percentage of them generates knowledge, either because they are already known or because they are not relevant to the user. To make the selection of these rules, they need to be assessed on their level of interest for the user in a specific context. Despite many attempts to give a formal definition on what makes a rule interesting, there is still no agreement. To [J Frawley et al., 1992], he identifies them as rules that are novel, useful and non trivial to compute, on the other hand to [Shen et al., 2002] a rule's interestingness is his probability added to his utility, while [Geng and Hamilton, 2006] have adopted a more broad definition stating 9 criteria that rules should meet to be considered interesting:

1. **Conciseness** : A pattern is concise if it contains relatively few attribute-value pairs, while a set of patterns is concise if it contains relatively few patterns. Being concise makes it easier for the pattern to be understood, remembered and incorporated into the user beliefs.

2. **Generality/Coverage** : A pattern is general if it covers a relatively large subset of a data-set, thus its more likely to be interesting [Agrawal and Srikant, 1994]. Generality and Conciseness tend to coincide since concise patterns tend to have greater coverage. Also, it tends to coincide with Reliability and conflict with Peculiarity.

3. **Reliability**: A pattern is reliable if the relationship described by the pattern occurs in a high percentage of applicable cases.

4. **Peculiarity** : A peculiar pattern, generated by outliers, is a pattern that is dissimilar to other discovered patterns. Since these patterns are usually unknown to the user, they can be interesting. Tends to coincide with Novelty.

5. **Diversity** : A pattern is diverse if its elements differ significantly from each other, while a set of patterns is diverse if the patterns in the set differ significantly from each other.

6. **Novelty** : A pattern is novel to someone if it was unknown and cannot be deduced from other patterns. Since no DM system represents everything a user knows or dont knows, novelty is identified by having the user to explicitly identify the pattern as novel [Sahar, 1999] or by noticing that the pattern cannot be deduced from and does not contradict previously discovered patterns.

7. **Surprisingness** : A pattern is surprising if it contradicts a person's existing knowledge or expectations [Silberschatz and Tuzhilin, 1996]. The difference between surprisingness and novelty is

that a novel pattern is new and not contradicted by any pattern already known to the user, while a surprising pattern contradicts the user's previous knowledge or expectations [Liu et al., 1997, Liu et al., 1999b, Silberschatz and Tuzhilin, 1995, Silberschatz and Tuzhilin, 1996].

8. **Utility** : A pattern is of utility if its use by a person contributes to reaching a goal. Different people may have divergent goals concerning the knowledge that can be extracted from a data-set.

9. **Actionability** : A pattern is actionable (or applicable) in some domain if it enables decision making about future actions in this domain. Considered by [Silberschatz and Tuzhilin, 1996] as a good aproximation for Surprisingness and vice versa.

Having defined the different criteria, the process to determine whether a pattern is interesting or not, starts by classifying each pattern as interesting or uninteresting using the above criteria. Then, a preference relation is chosen, so that one pattern is represented instead of other, i.e. produces a partial ordering. Finally, the pattern are ranked based on the chosen criteria. Thus, using interesting measures facilitates a general and practical approach to automatically identifying interesting patterns [Geng and Hamilton, 2006].

2.3.1 Interestingness Measures

Given an AR, his interesting value can be determined using up to three types of measures. Objective ones, which can be divided in: Probabilistic employing the Generality and Reliability criteria, and Rules Form using Peculiarity, Surprisingness and Conciseness [Geng and Hamilton, 2006]. The most commonly used Objective measures used to assess the strength of an AR are the Probabilistic ones, in particular the Support, Confidence and Lift [Tan et al., 2005]. Their importance comes as they are many times the basis for new measures but also because they help reduce rules, especially the ones poorly correlated. However, objective measures don't take into account the context of the domain of application or the goals and background knowledge of the user. Subjective and semantics-based measures incorporate the user's background knowledge and goals, respectively, and are suitable both for more experienced users and interactive data mining. Subjective measures rely on Surprisingness and Novelty, while Semantic ones use Utility and Actionability to help identify the most interesting rules.

2.3.1.1 Objective Measures

Objective measures which are derived from statistics and information theory to rank the numerical or structural properties of a rule, depend only on raw data (i.e. no previous knowledge is needed) [Geng and Hamilton, 2006]. Traditionally, a rule's interest assessment is determined using objective measures such as support, confidence and lift [Vreeken and Tatti, 2014, Silberschatz and Tuzhilin, 1996].

Support : Represents the generality of a rule. Shows how often a rule, with respect to a set of transactions T, can be applied to a dataset. Rules that have low support typically occur by fortuity and often are uninteresting from a business perspective.

$$\mathrm{Supp}(X \Rightarrow Y) = \frac{|\{t \in T; X \subseteq t\}|}{|T|} \tag{2.4}$$

Confidence : Represents the reliability of a rule. Shows how often the AR has been found to be true, i.e., the reliability of the association made by the rule confidence thus estimates the rule's conditional probability.

$$\mathrm{Conf}(X \Rightarrow Y) = \frac{\mathrm{supp(X \cup Y)}}{\mathrm{supp(X)}} \tag{2.5}$$

However, both have well known flaws when in specific situations: Support has trouble with rare items as infrequent ones fail to meet the Minimum Support (MINSUP) and thus are ignored [Liu et al., 1999a]; other issue is the fact that it favors short item-sets [Seno and Karypis, 2005]. Confidence also has problems, especially since it ignores the support of the itemset in the rule's consequent [Aggarwal and Yu, 1998, Silverstein et al., 1998]. Thus, other measures to increase the chance of successfully identifying interesting rules are needed. Since its impractical to list all Interesting Measure (IM) available, I have selected three that complement the Support and Confidence measures.

Lift : Introduced by [Brin et al., 1997], shows to what extent X and Y are dependent on one another. Lift is a symmetric measure with respect to antecedent and consequent of a rule, that allows to measure the co-occurrence (not implication) in order to retrieve rare important interest rules pruned by the user-defined support and confidence thresholds [Azevedo and Alípio, 2007].

$$\mathrm{Lift}(X \Rightarrow Y) = \frac{\mathrm{conf(X \Rightarrow Y)}}{\mathrm{supp(Y)}} \tag{2.6}$$

A lift value of close to 1 indicates that X and Y are independent and the rule is not interesting.

Conviction : Also introduced by [Brin et al., 1997], overcomes the insensitivity of Lift to rule direction, i.e. $X \Rightarrow Y =$ and $Y \Rightarrow X$, when measuring the degree of implication of a rule. Also, unlike confidence, the support of both antecedent and consequent parts of a rule are taken in consideration. An interesting observation is that Conviction is monotone in Confidence and Lift [Azevedo and Alípio, 2007, Manimaran and Velmurugan, 2015].

$$\mathrm{Conviction}(X \rightarrow Y) = \frac{1 - \mathrm{Support(Y)}}{1 - \mathrm{Confidence(X \rightarrow Y)}} \tag{2.7}$$

Conviction ranges from 0.5 to ∞, where 1 indicates X and Y are independent, thus uninteresting rules, and values far from 1 that rules are interesting.

Chi-Square : Shows the degree of dependence between variable X and Y by comparing the observed frequencies and the corresponding expected frequencies. It requires the creation of two contingency tables(observed and expected), each having all four possible combinations between X and Y, as shown in table 2.1 and table 2.2, with n being the total number of samples.

Table 2.1: Contingency table : observed frequency

	Y	$\bar{Y}$
X	$nP(X \cap Y)$	$nP(X \cap \bar{Y})$
$\bar{X}$	$nP(\bar{X} \cap Y)$	$nP(\bar{X} \cap \bar{Y})$

Table 2.2: Contingency table : expected frequency

	Y	$\bar{Y}$
X	$nP(X)P(Y)$	$nP(X)(1-P(Y))$
$\bar{X}$	$n(1-P(X))P(Y)$	$n(1-P(X))(1-P(Y))$

Let k be the number of categorical attributes, ef_j and of_j represent the absolute values of the expected and observed frequency in category j. Then, χ^2 can be defined as:

$$\chi^2 = \sum_{j=1}^{k} \frac{(ef_j - of_j)^2}{\mathrm{of_j}} \tag{2.8}$$

2.3.2 Unexpectedness and Novelty Measures

Subjective Measures (SM) consider data and user knowledge of these data and are based on notions of unexpectedness (i.e., is interesting if it surprises the user) and actionability (i.e., is interesting if the user can use it to make a decision and obtain some advantage) [Silberschatz and Tuzhilin, 1996] but also novelty criteria [Boettcher et al., 2009]. They are recommended when the background of users varies, users interest varies and the background knowledge of users evolve. Contrary to the measures described in Section 2.3.1.1, subjective ones cannot be represented by simple mathematical formulas as user knowledge can be expressed in several formats. Instead, they are incorporated in the mining process [Geng and Hamilton, 2006]. Three approaches can be distinguished :

- **Filter Interesting Patterns from Mined Results** : A formal specification of the user knowledge is given and after obtaining the DM results, the unexpected patterns are presented. [Silberschatz and Tuzhilin, 1996] proposed a framework for defining an IM for patterns using Bayesian approach, which related unexpectedness to a belief system. In this system, a belief can be classified as Hard

and Soft. A Hard belief is a constraint that cannot be changed with new evidence (mined rule), even if the evidence contradicts hard beliefs, a mistake is assumed to have been made when acquiring the evidence. A Soft belief is one that the user is willing to change as new patterns are discovered.

- **Eliminating Uninteresting Patterns** : [Sahar, 1999] proposed an process that removes uninteresting rules, rather than selecting interesting ones, based on user feedback. The process consists in 3 steps, which are iterated until the rule-set becomes empty :

  1. The best candidate rule is selected as the rule with exactly one condition attribute in the antecedent and exactly one consequence attribute in the consequent that has the largest cover list. The cover list of a rule R is all the mined rules that contain the condition and consequence of R.
  2. The best candidate rule is presented to the user for classification into one of four categories: not-true-not-interesting, not-true-interesting,true-not-interesting, and true-and-interesting. If the best candidate rule R is not-true-not-interesting or true-not-interesting, the system removes it and its cover list. If the rule is not-true-interesting, the system removes this rule as well as all the rules in its cover list that have the same antecedent, and keeps all the rules in its cover list that have more specific antecedents
  3. Finally, if the rule survives, then it is true-interesting and the system keeps it.

  Useful when the user does not want to explicitly represent knowledge about the domain

- **Constraining the Search Space** : User specifications are used as constraints during DM process to reduce the search space and consequently reduce the number of results. [Padmanabhan and Tuzhilin, 1998] proposed a method to narrow down the mining space on the basis of the user's expectations. In this method, no IM is defined. Here, the user's beliefs are represented in the same format as mined rules. Only surprising rules, that is, rules that contradict existing beliefs, are mined. Useful when the user knows what kind of patterns he or she wants to confirm or contradict.

2.3.3 Semantic Measures

Semantic measures regard the semantic and explanations of patterns. Since semantic measures include domain knowledge from users, they can also be considered as a sub-type of objective measures [Yao and Hamilton, 2006]. They are based on :

- **Utility Based Measures** Relevant semantics are the utilities of the patterns in the domain (most common). Contrary to SM where domain knowledge is about data represented similarly to the discovered patterns, for semantic measures do not relate user's knowledge regarding data. Rather,

it represents a utility function that considers the statistical aspects of the raw data and the utility of the mined patterns, in order to reflect user's goals. This is especially suited for decision-making problems in real world applications.

- **Actionability** Actionable patterns can help user's take decisions to their advantage. To identify these patterns [Liu et al., 1997] proposed a method where the user supplies patterns in the form of fuzzy rules, representing both possible actions and the situations in which they are likely to be taken. Then, the system match each discovered pattern against the fuzzy rules and ranks them. Those with the highest value of matching are the ones chosen to be selected.

2.3.4 Retrospective Cohort Studies

Retrospective Cohort Studies have been widely accepted for identifying causal links in health, medical and social studies. Researchers travel to a point in time before the outcomes of interest (e.g., hypertension) have developed, and try to establish a relation to the outcome based on the status of being exposed to a potential cause factor (e.g., eat salty food). The process begins by hypothesizing a potential causal rule, followed by the creation of an exposure and a non-exposure or control group of individuals to a suspected risk factor. While both groups differ on the exposure to the risk factor, they are alike in other aspects (e.g., age, gender, location) and are followed to observe the occurrence of the outcome. The effect of the exposure factor (Odds Ratio) is then determined by comparing the difference between the exposure and control groups.

As previously stated in Section 2, in ARM one of the principal problems is the vast amount of uninteresting rules generated. Since in DM the source of information are historical records, [Li et al., 2015] proposed a statistically sound and computational efficient causal discovery method for causal relationship exploration based on these studies.

Let us consider a data-set D, and the association rule $p \Rightarrow z$ as an hypothesis. Let p be the exposure and z the response variables with c representing the set of control variables. The process begins by choosing a record containing p, then other not containing p while z is blinded and both have matched values for c (matched pair). Then, each one is removed from D and attributed to the corresponding group and the process repeats until no more matched pairs can be found (random selection). The result, fair data-set, is the maximal sub data-set of D that contains only matched record pairs. There are four possibilities for a matched pair: both records contain z (n_{11}), neither contain z (n_{22}), exposure group containing z and non-exposure does not (n_{12}), and vice versa (n_{21}) as shown in table 2.3.

Now, Odds Ratio of a rule $p \Rightarrow z$ on a fair data-set D_f can be calculated as:

$$\mathrm{OddsRatio_{D_f}(p \Rightarrow z)} = \frac{\mathrm{n_{12}}}{\mathrm{n_{21}}} \tag{2.9}$$

		$P=0$	
		z	$\neg z$
$P=1$	z	n_{11}	n_{12}
	$\neg z$	n_{21}	n_{22}

Table 2.3: Contingency table : exposure groups

This way, when the odds ratio of an association rule on its fair data-set is significantly greater than 1, it means that a change of the response variable is a consequence of the exposure variable, and thus indicative of a causal rule.

2.3.5 Frameworks

Both Objective and Subjective measures have flaws which prevents them from being used alone in many real-world applications i.e. no single measure is superior to all others or suitable for all applications. Objective ones are independent of the domain in which the data mining process is performed, they do not take into consideration the knowledge and goals of the specialists when searching for interesting rules. On the other hand, subjective ones requires a user to know in advance what he is looking for [Rezende et al., 2009]. Also, since they treat the domain knowledge as static, there is the probability of identifying rules as interesting based on outdated knowledge [Boettcher et al., 2009]. However they are still important, as objective ones give a first impression at what was discovered, setting the starting point for further exploration using subjective ones. Therefore, interestingness should be assessed using both measures. Next, I will present two frameworks that take both these measures into consideration and can be suitable in the context of this research.

2.3.5.1 Rule Changing + Relevance Feedback

Proposed by [Boettcher et al., 2009], this powerful and intuitive framework combines objective and SM of interestingness as well as user feedback in order to find the most interesting rules from the set. It generates potentially interesting time-dependent features for ARs during post-mining, which are then combined with the rule's textual description using relevance feedback methods from information retrieval [Geng and Hamilton, 2006].

Leveraging on the notion of change, rules that change over time may suggest surprising changes present in the data generating process, thus requiring intervention. For instance, rules with a dipping trend in confidence point that it might disappear in the future, while those with a rising trend might indicate the appearance of a rule. Contrarily, stable rules often represent invariant properties of the data generating process and thus are either already known and should not be presented again. The framework consists in 4 phases:

1. **Rule Discovery** AR have to be discovered and their histories efficiently stored, managed and maintained. If histories with a sufficient length are available, the next task is straightforward and constitutes the core component of rule change mining.

2. **Change Analysis** Since a history is derived for each rule, the rule quantity problem also affects rule change mining: it has to deal with a vast number of histories and thus it is likely that many change patterns will be detected. Furthermore, there is also a quality problem: not all of the detected change patterns are equally interesting to a user and the most interesting are hidden among my irrelevant ones.

3. **Objective Interestingness** An initial interestingness ranking for ARs proves to be helpful in providing a user with a first overview over the discovered rules and their changes.

4. **Subjective Interestingness** User feedback about rules and histories seen thus far should be collected, analysed and used to obtained new interestingness ranking.

2.3.5.2 Data Driven + User Driven

Proposed by [Rezende et al., 2009], this framework combines the use of data-driven and user-driven measures, focusing strongly on the interaction with the expert user. The framework consists in 4 phases:

1. **Objective Analysis** The aim is to use objective measures in the selection of a subset of the rules generated by an extraction algorithm. Then the subset can be evaluated by the user. The selection of a rule subset is done by using rule set querying and objective measures. The rule set querying is defined by the user if there is the wish to analyze rules that contain certain items. After the distribution analysis of objective measure values a cut point is set to select a subset of rules. The cut point filters the focus rule set for each measure. The union/intersection of the subsets defined by each measure forms the subset of PIR.

2. **User Knowledge & Interest Extraction** Can be seen as an interview with the user to evaluate rules from the PIR subset. In order to optimize the evaluation, rules are ordered according to the itemset length since they are simpler to comprehend. For each rule from the PIR subset, the user has to indicate one or more evaluation options, classifying the knowledge represented by a rule (unexpected, useful, obvious, previous, irrelevant) considering analysis goals.

3. **Evaluation Processing** In focus rule set, defined in objective analysis, irrelevant rules are eliminated and SM are calculated. Every time a rule is classified as irrelevant knowledge by the user, means that the user is not interested in the relation among the items, therefore all similar rules from the focus rule set are eliminated.

4. **Subjective Analysis** The user can explore the resultant focus rule set using the chosen SM as a guide. The user accesses the rules according to each measure and considers each evaluated rule. This exploration should be carried out according to the goals of the user during the analysis. By browsing the focus rule set, the user identifies his rules of interest. Thus, at the end of the analysis, the user will have a set of rules which were considered interesting.

2.4 Summary

This chapter began by introducing the concepts of ARM and SPM to this work. The original approach to mine AR, although simple, had performance problems due to the exponential growth of generated rules. The algorithm developed by [Agrawal and Srikant, 1994] named Apriori, solved this problem by pruning candidates that do not meet the minimum support required. However, this still required the input file to be read in each iteration, thus calculating the pairs to be counted. To overcome these problems [Han et al., 2000] proposed a different approach that leverages on a tree-like structure for paths. Regarding the SPM, [Pei et al., 2001] proposed an algorithm named PrefixSpan to mine FS based on database projections. Due to these projections have to be made in each level, [Pei et al., 2001] also proposed two extensions that use Bi-Level and Pseudo projections to avoid doing so. Next, we looked at different approaches on how a rule's interest factor can be evaluated using objective, subjective and semantic measures. Objective measures are derived from statistics and information theory to rank rules, depending only on raw data using measures such as Support, Confidence and Lift. Subjective ones are based on concepts of Unexpectedness, Actionability and Novelty leveraging on the work context and the knowledge of the user. Since both types of measures contained flaws, [Boettcher et al., 2009] proposed a framework based around the notion of change, that combines the two types of measures as well as user feedback, to help identify the PIR. Similarly, [Rezende et al., 2009] also proposed a framework based on the two types of measures but focused more on user knowledge. Finally, since causal relations in medical context are most likely PIR, Retrospective Cohort Studies were introduced and the concept of Odds Ratio was explored as a way to identify causal rules and therefore PIR.

In the next chapter, novel advanced techniques to find relevant patterns are discussed, including those specifically created to perform DM in electronic medical prescription databases.

3. Related Work

This chapter presents relevant previous studies in the context of my work. Section 3.1 explores advanced techniques that extend the capabilities of SPM to identify new types of structures. Techniques specifically designed to mine medical prescription databases or other types of electronic health records are also reviewed in Section 3.2. Finally, Section 3.3 provides a brief summary and comparison about the techniques that were reviewed.

3.1 Advanced Approaches for Mining Sequential Data

Traditionally, sequential pattern mining techniques focus on finding relevant patterns in ordered sequences of events. However some challenges still remain when mining event databases, such as defining boundaries and process instances to search for local patterns [Leemans and van der Aalst, 2015, Tax et al., 2016], using event abstractions to facilitate model comprehension [Mannhardt et al., 2018, Chapela-Campa et al., 2017], considering contextual information [Boytcheva et al., 2017] and understanding the relations between patterns [Lu et al., 2008]. Next, I present techniques, related to process mining (i.e., a combination of data mining and process modelling) that address these challenges.

3.1.1 Post Sequential Pattern Mining & ConSP-Graph

[Lu et al., 2008] described a method to discover complex structural patterns hidden behind sequences, in order to represent relations between sequential patterns. This method leverages on traditional SPM techniques to generate sequential patterns, and then continues processing them to discover Structural Relation Patterns (SRP).

The main idea behind the method of [Lu et al., 2008] is to search for sequential patterns supported by data sequences, which can then be used to determine SRP patterns, and subsequently to determine the corresponding maximal set. Since sequential patterns supported by data sequences can be viewed as a transaction, the problem of mining SRP, in particular concurrent patterns, satisfying a minimum confidence is similar to mining frequent item-sets under minimum support.

One particular case of SRP are concurrent patterns. Concurrence was defined by [Lu et al., 2010] as the fraction of data sequences that contain all of the sequential patterns. Let us assume that SDB is a sequence database, i.e. it generates unique values for each record. Assume also that $\{sp_1, sp_2, ..., sp_m\}$

is the set of m sequential patterns mined under min-support and that they are not contained in each other. Then, a pattern concurrency can be defined as:

$$\text{concurrence}(sp_1, sp_2, ..., sp_i) = \frac{|\{C : \forall_i (i = 1, 2, ..., k) sp_i \angle C \in \text{SDB}\}|}{|\text{SDB}|} \tag{3.1}$$

In the previous equation, $sp_i \angle C$ indicates the sequential pattern sp_i is contained in data sequence C. Concurrent Sequential Patterns (CSP) are sequential patterns whose concurrency is above the min-confidence threshold. CSP are represented by $\text{ConSP}_\text{k} = [sp_1 + sp_2 + ... + sp_k]$ where k is the number of sequences and $+$ denotes the concurrent relationship.

To mine these patterns, we begin by identifying the sequential patterns who have support in the sequential patterns occurring together. Notice that ConSP_k assures that the patterns occur above a certain threshold, although it does present its minimal representation as further relations have not yet been explored. These patterns can be viewed as transactions and our problem of finding CSP satisfying min-confidence can be solved using techniques to mine frequent item-sets satisfying min-support. The resulting patterns must then be simplified to ensure they are not contained in any other concurrent pattern. The simplified set of maximal concurrent patterns can be obtained by deleting concurrent patterns which are contained by other concurrent patterns, and/or by deleting sequential patterns which are contained by other sequential patterns when mining for frequent item-sets.

To explore the inherent relationship among sequential patterns, in particular CSP, a graphical representation named Concurrent Sequential Patterns Graph (ConSP-Graph) was developed [Lu et al., 2010]. This graph is defined by the septuple $(V, E, S, F, S', F', \sigma)$, where V is a nonempty set of nodes, E is a set of directed edges, S is a set of start nodes, F is a set of final nodes, S' is a set of synchronizer nodes, F' is a set of fork nodes, and σ is a function from a set of directed edges to a set of pairs of nodes. The process of constructing the graph involves 5 steps:

1. **Initialization**: An initial overall model is constructed by composing the direct graphs $G(\beta_i)$ representing each a sequential pattern β_i. A transitional model is also initialized as $G' = \emptyset$.

2. **Refinement**: For all pairs of $G(\beta_i)$ and $G(\beta_j)$, with $i \neq j$, refine the overall model by finding each occurrence of a common prefix and/or postfix. When a pair of graphs share a prefix/postfix jump to Step 3 of the algorithm, otherwise continue through each remaining pair of graphs in G until this cycle is complete, and then go to Step 4.

3. **Combination**: merge two sharing prefix/postfix graphs, $G(\beta_i)$ and $G(\beta_j)$, into a new one and put it in transitional model G'. Return to Step 2.

4. **Deletion**: remove graphs used in merging from G and insert newly created merged graphs into G', which now form a new overall model G.

5. **Iteration**: reiterate through Steps 2-4 until no more merges can be made. The final result G' is the ConSP-Graph.

The resulting graph ensures that (1) for any node, the subsequent paths of it cannot be the same and the ancestor paths of it cannot be the same either,and (2) for each pair of different nodes, if they have the same value, there must be different ancestor paths and subsequent paths for them. Despite bringing connectivity and structure to patterns, ConSP-Graph was naturally found prone to over-fitting problems [Tax et al., 2016].

3.1.2 Local Process Model

Mining Local Process Model (LPM) enables the discovery of frequent behavioural patterns (e.g., sequential composition, concurrency, choice and loops) in event logs that are too unstructured for regular process mining techniques [Tax et al., 2016]. It focuses on subsets of process activities, describing their inner behavioral patterns.

Since this method leverages on process trees to search LPM, we must first define this concept. A process tree is able to model sound processes (i.e., deadlock and live-lock free) and is represented by a tree structure, where the leaf nodes designate activities and the non leaf nodes control-flow operators. These include a loop operator ($\circlearrowleft$) where the first child is the *do* part and the second child the *redo*. The *do* part will always be executed at least once and is both the start and endpoint of the loop. We also have a sequence operator ($\rightarrow$), where the left child is executed prior to the right one. The exclusive choice operator (X) indicates that only one of the childs will be executed, whereas the concurrency operator ($\wedge$) has both children executed in parallel. The set of activity labels A' is expanded by a silent activity represented by τ. This activity cannot be observed and its purpose is to model processes where an activity can be skipped under specific circumstances.

Let $A \in A'$ be a set of label activities with $\tau \notin A^{'}.\oplus = \{\rightarrow, X, \wedge, \circlearrowleft\}$. A process tree exists according to the following conditions:

- If $a \in A' \cup \{\tau\}$, then Q is a process tree.
- If Q_1, Q_2 are process trees, and $\oplus \in \bigoplus$, then $Q = \oplus(Q1, Q2)$.

Process trees can be optimized by restricting the expansion of a leaf node that has a symmetrical operator as parent with the same symmetrical operator only to the rightmost child, preventing unnecessary computation. Let LPM LN represent behavior over $A^{'}$ accepting language $\pounds(LN)$, and let L denote the corresponding alphabet. The LPM method consists of 4 steps:

1. **Generation**: An initial set of candidate LPM, in process tree format and with one leaf for each activity $a \in L$, is generated and represented as CM_1 (i.e., considering $i = 1$). A set to store the selected LPM is also created, i.e., $\mathrm{SM} = \emptyset$.

2. **Evaluation**: An assessment is made on the process trees in $\mathrm{CM_i}$ based on defined quality criteria (e.g., support and/or confidence).

3. **Selection**: The assessed trees that comply with the defined quality criteria $\mathrm{SCM_i} \subseteq \mathrm{CM_i}$ are added to $\mathrm{SM} = \mathrm{SM} \cup \mathrm{SCM_i}$. When $\mathrm{SCM_i} = \emptyset$ or $i \geq \mathrm{max_iterations}$ the procedure stops.

4. **Expansion**: $\mathrm{SCM_i}$ is expanded into a set of bigger candidate process trees, $\mathrm{CM_{i+1}}$. We then return to Step 2 with the generated candidate set $\mathrm{CM_{i+1}}$.

Since the computational time used in finding LPM grows rapidly with the number of activities present in the event log, quality dimensions have been defined to limit the search space, thus increasing the associated efficiency. Using thresholds and weights on these dimensions, undesired generated models are pruned using Apriori monotonicity properties. Dimensions include:

- **Support**: linked with the number of fragments in the event log that can be considered an instance of the LPM in assessment.
- **Confidence**: if associated to an event, confidence is the proportion of events of a type present in the log that fit in LPM. If associated to LPM, confidence is the harmonic mean of individual confidence scores of the event types presented on it.
- **Language Fit**: the ratio of behavior permitted by LPM that is observed in the event log. Allowing for too much behavior can lead to over-generalization, thus failing to properly describe the log.
- **Determinism**: deterministic LPM have bigger predictive value regarding future behavior.
- **Coverage**: the ratio of the number of events in the log after projecting the event log on the labels of LPM to the number of all events in the log.

Despite using process trees to identify LPM, this approach does not suffer from over-fitting like PSPM & ConSP, since its does not merge all patterns into one graph, instead returning a set of patterns. In testing, this method was capable of mining noisy data and find long-term dependencies.

3.1.3 Frequent Episode Mining

Frequent Episode Mining is a technique based on the discovery of frequent item-sets that explores the notion of process instances to automatically adjust the boundaries of local processes [Leemans and van der Aalst, 2015]. Episodes are collections of partially ordered events for a consecutive and fixed time intervals in a sequence. Since events are associated with cases, this technique leverages on this knowledge to find frequently occurring episodes (i.e., local patterns) in temporal databases (e.g., event logs) unlike SPM which applies to sequence databases.

Let $\mathcal{A}$ be the alphabet of activities. A trace is a list $T = \langle a_1, ..., a_n \rangle$ of activities $a_i \in \mathcal{A}$ occurring at time index i relative to other activities in T. An event log $L = [t_1, ..., t_m]$ is a multiset of traces t_i. Each trace corresponds to an execution of a process, i.e. a process instance.

A partial ordered collection of events is called an episode and it is represented by a triple where V is set of nodes representing events, $\leq$ is a partial order on V, and g is the node labeling function. If $|V| \leq 1$ then we have an empty episode. When $\leq \emptyset$ we call α a parallel episode.

$$\alpha = (V, \leq, g) \tag{3.2}$$

An episode $\beta = (V^{'}, \leq^{'}, g^{'})$ is a sub-episode of $\alpha = (V, \leq, g)$, denoted $\beta \preceq \alpha$, iff there is an injective mapping $f : V^{'} \mapsto V$ such that:

$$(\forall_v \in V^{'} : g^{'}(v) = g(f(v))) \wedge (\forall_{v,w} \in V^{'} \wedge v \leq^{'} w : f(v) \leq f(w)) \tag{3.3}$$

An episode β equals an episode α, denoted $\beta = \alpha$ iff $\beta \preceq \alpha \wedge \alpha \preceq \beta$. If $\beta \preceq \alpha \wedge \beta \neq \alpha$ then β is called strict subepisode and is represented by $\beta \prec \alpha$.

The construction of a new episode from two previous episodes α and β is represented by $\gamma = \alpha \oplus \beta$, where $\alpha \oplus \beta$ is the smallest γ such that $\alpha \preceq \gamma$ and $\beta \preceq \gamma$. Two episodes $\alpha = (V, \leq, g)$ and $\beta = (V^{'}, \leq^{'}, g^{'})$ can be merged to construct a new episode $\gamma = (V^{*}, \leq^{*}, g^{*})$. An episode $\alpha = (V, \leq, g)$ occurs in an event trace $T = \langle a_1, ..., a_n \rangle$, denoted $\alpha \sqsubseteq T$, iff there exists an injective mapping $h : V \mapsto 1, ..., n$ such that:

$$(\forall_v \in V : g(v) = A_{h(v)}) \wedge (\forall_{v,w} \in V \wedge v \leq w : h(v) \leq h(w)) \tag{3.4}$$

The frequency $\mathrm{freq}(\alpha)$ of an episode α in an event log $L = [t_1, ..., t_m]$ corresponds to rate at which the episode appears in the log:

$$\mathrm{freq}(\alpha) = \frac{|[t_i | t_i \in L \wedge \alpha \sqsubseteq t_i]|}{|L|} \tag{3.5}$$

Let minFreq be the minimum frequency threshold. An episode α is frequent iff $\mathrm{freq}(\alpha) \geq \mathrm{minFreq}$. Defined similarly, the activity frequency $ActFreq(a)$ of an activity $a \in \mathcal{A}$ in an event log $L = [T_1, ..., T_m]$ is the ratio at which the activity appears in the log

Given an episode $\alpha = (V, \leq, g)$ occurring in an event trace $T = \langle a_1, ..., a_n \rangle$, as indicated by the event to trace map $h : V \mapsto 1, ..., n$, the trace distance regarding an episode in an event trace is defined as:

$$\mathrm{traceDist}(\alpha, T) = \max\{h_v | v \in V\} - \min\{h(v) | v \in V\} \tag{3.6}$$

Since in LPM we are only interested in a partial ordering of events that occur relatively close in time, an episode α is accepted in trace t, regarding trace distance interval, iff $minTraceDist \leq \mathrm{traceDist}(\alpha, t) \leq$

$maxTraceDist$.

A useful concept to filter generated rules that are trivial is the concept of magnitude. Let $\mathrm{size}(\alpha)$ be the graph size of an episode α which can be calculated as the sum of the nodes and edges in the transitive reduction of the episode. The magnitude of an episode rule $\beta \Rightarrow \alpha$ represents how much an episode α adds to episode β with values ranging from 0 to 1 and is defined as:

$$\mathrm{mag}(\beta \Rightarrow \alpha) = \frac{\mathrm{size}(\beta)}{\mathrm{size}(\alpha)} \tag{3.7}$$

Very low/high magnitude numbers are indicative of trivial episode rules.

The following properties regarding episodes are also used in the algorithm:

- If an episode α is frequent in an event log L, then all subepisodes β with $\beta \preceq \alpha$ are also frequent in L.
- If an episode rule $\beta \Rightarrow \alpha$ is valid on an event log L, then for all episodes $\beta^{'}$ with $\beta \prec \beta^{'} \prec \alpha$ the event rule $\beta^{'} \Rightarrow \alpha$ is also valid in L.

The episode mining algorithm consists of 5 steps:

1. **Frequent Episode Discovery**: The first step divides itself in two phases: one focuses on discovering parallel episodes (i.e. nodes only) while the other focuses on partial orders (i.e. adding the edges). The result is a set of frequent episodes.

2. **Episode Candidate Generation**: This step is based on the Apriori algorithm. For the parallel phase, we have that F_l contains frequent episodes with l nodes and no edges. A candidate episode γ will have $l+1$ nodes, resulting from episodes α and β that overlap the first $l-1$ nodes. For the partial ordering phase the process is the same but applied to edges, the only difference being that episodes α and β, besides overlapping the first $l-1$ edges must also have the same set of nodes.

3. **Frequent Episode Recognition**: Regardless of phases, candidate episodes are assessed to see if they are frequent. To check if a candidate episode α is frequent, we check if $\mathrm{freq}(\alpha) \geq \mathrm{minFreq}$. To check if an episode α appear in a trace $T = \langle a_1, ..., a_n \rangle$ we need to check the existence of a mapping $h : \alpha.V \Rightarrow \{1, ..., n\}$, which can be done ensuring two things:

   - Each node $v \in \alpha.V$ has a unique witness in trace T.
   - Mapping h respects the partial order indicated by $\alpha. \leq$.

   In the end, a set of frequent episodes is returned.

4. **Pruning**: To reduce the number of uninteresting episodes, thus making the algorithm more resistant to logs with infrequent activities, the activity alphabet $\mathcal{A}$ can be replaced by $\mathcal{A}^* \subseteq \mathcal{A}$, with

$(\forall A \in \mathcal{A} : ActFreq(A) \geq minActFreq)$. Also, we can prune episodes based on the trace distance interval.

5. **Episode Rule Discovery**: For all the frequent episodes α, we consider all the frequent subepisodes β with $\beta \prec \alpha$ for the episode rule $\beta \Rightarrow \alpha$.

If a frequent episode β is created by merging frequent episodes γ and ϵ, then β is the child of γ and ϵ. Similarly, γ and ϵ are his parents. We can travel from an episode α along the discovery parents of α. When we find a parent β with $\beta \prec \alpha$, we can also consider the parents and children of β and on the property of validity regarding an episode we cannot apply pruning in either direction of the parent-child relation.

Through experiments assessing the performance of the algorithm, the authors showed it is fast for a low number of episodes (<500). However, setting the optimal values of the thresholds $minFreq$, $minActFreq$ and $maxTraceDistance$ is challenging.

3.1.4 Guided Process Discovery

Since low-level events stored in event logs often do not correspond to high level activities that are intuitive to stakeholders, a new method called Guided Process Discovery (GPD) was developed using event abstractions based on process domain knowledge to produce easy to understand models to stakeholders [Mannhardt et al., 2018].

Let us assume that multiple low-level events grouped together represent the execution of a high level activity. The GPD procedure starts with a low-level event log where activities are recorded. The execution of a process instance generates a sequence of events, each with 3 attributes: name of the activity causing event, time of occurrence, and activity instance (i.e., an identifier linking multiple events to the same activity). Then, activity patterns capturing the domain knowledge and describing low-level event interactions are identified. Using these patterns, an abstraction model is created that is then aligned with the initial event log, resulting in an aligned low-level event log. Using alignment mapping, this log is then abstracted to an high-level event log at the desired abstraction. Finally, an high-level process model is created.

To check the quality of the resulting model, activities in the abstraction model are replaced by their activity patterns and then the expanded model is compared against the initial event log, with quality measures being used to assess the reliability of the abstraction.

Each of steps involved in the algorithm are described next in more detail:

1. **Identification and encoding of activity patterns**: An activity pattern indicates the sequence order necessary to run the high-level activity on a low-level event, as well as its name. The labeled process model identifies the expected events to be seen in the event log for one instance of the

associated activity. However, activity patterns do not guarantee an exact representation of the activity, since they can be displayed in multiple ways in the event log.

2. **Manual patterns**: These patterns are created based on domain knowledge regarding high-level activities of the process. These include:
   - Expert Knowledge, which encode assumptions on the system.
   - Process Questions, which can be used as a source for activity patterns.
   - Standard Models, which are independent of the concrete domain. Patterns based on standard models appear in processes across all domains.
3. **Discovered patterns**: These patterns are automatically discovered from the low-level event log and they include:
   - Local Behavior Patterns, which do not capture the behavior of complete traces, but instead on event subsets. They are similar to the LPM technique.
   - Decomposed Behavior Pattern, which leverage on decomposition approaches to obtain activity patterns that display parts of the observed behavior.
   - Data Attributes, which explore data on the event log hierarchical structure.
4. **Activity Pattern Composition**: An abstraction model, displaying overall behavior from the execution of all high-level activities in a single instance, is made through composition of captured behavior in activity patterns of the associated instance. Compositions include, but are not limited to, sequence, choice, parallel, interleaving and repetition.
5. **Event Log and Abstraction Model Alignment**: An high-level event log is created using an alignment of low-level event log entries with the abstraction model. The need for alignment comes from the fact that noise exists in event logs, and therefore not all low-level events can be mapped into high-level activities. Once modeled, quality measures are computed regarding how the entire low-level event log (i.e., global matching error), and how each identified high-level activity (i.e., local matching error) matches the behavior imposed by the abstraction model.
6. **High Level Process Model**: Using any process discovery techniques that can exploit the fact that information on the life-cycle activities are contained in the abstract event log (e.g., Inductive Miner), allows the discovery of a process model based on the abstracted high level activities.
7. **Expansion and Validation**: To evaluate the quality of the model generated in Step 6, we can substitute every high-level activity by the associated activity pattern. The resulting expanded model describes the behavior of the previous model in terms of low-level events. Then, the model is put against an event log to assess its quality.

Tests showed that GPD can deal with noisy data, reoccurring and concurrent behavior, and shared functionality. The resulting models not only make a good representation of events logs, but they are also capable of answering process questions and are intuitive to stakeholders. However, the alignment process becomes very expensive for traces with more than 250 events.

3.1.5 WoMine

An algorithm, based on Apriori, called WoMine was proposed to identify and retrieve frequently executed structures with sequences, selections, parallels and loops on already discovered process models [Chapela-Campa et al., 2017]. One key feature of WoMine is that it can detect frequent patterns with all types of structures, including n-length cycles. The method is also robust regarding the quality of the mined models.

Regarding to patterns, they are sub-graphs of the process model that represents the behavior of parts of the process. For each task α in the pattern, its inputs, $\mathrm{I}^{'(\alpha)}$, must be a subset of $\mathrm{I}(\alpha)$ in the model it belongs to, and the outputs, $\mathrm{O}^{'(\alpha)}$, must also be a subset of $O(\alpha)$ in the model. This ensures that a pattern has not an incomplete parallel connection (i.e., the number of input choices of $\alpha > 1$). A Single Pattern (SP) is a pattern whose behavior can be executed entirely in one instance. If a task has a selection, then it must be able to execute each path in the same instance.

The objective of this algorithm is to find sub-graphs from a given process model that are executed in a percentage of traces above a given threshold. Let us assume that given a function f and a language $\mathcal{L}$, Minimal Pattern (MP) x is the smallest pattern with respect to the set inclusion in $\mathcal{L}$ satisfying the property $\mathrm{f(x)}$). The WoMine algorithm starts by initializing the frequent arcs of the candidate arc set $A^{<}$ and the frequent MP to measure their frequency. Then, these frequent arcs and the frequent M-Patterns will be used to expand other patterns with them by (1) adding frequent MP that are not in the current pattern, and (2) adding frequent arcs of $A^{<}$ to each of the current patterns. The result set is then pruned, leveraging on the anti-monotonicity property of support, removing patterns that failed to meet the frequency thresholds and redundant pattern (i.e., patterns whose behavior is contained in another one). To check the compliance of a trace τ regarding an SP belonging to the process model, the trace is compliant with SP, SP $\vdash \tau$, when the execution of the trace in the process model contains the execution of the pattern, i.e. all arcs and tasks of SP are executed in a correct order and each task fires the execution of its output in the pattern.

Tests showed that WoMine is a robust algorithm, as it extracts patterns (including patterns with loops, choices, parallels and sequences) from a given model but measures the frequency with the log. This allows it to successfully deal with low-fitness and high-generalization models. Furthermore, WoMine always returns the correct frequent patterns, even when other techniques fail to do so.

3.2 Mining Prescriptions and other Health Record Databases

Since early 2000's, health-care organizations transitioned from paper records to Electronic Medical Records (EMR), which led to huge amounts of data being collected in clinical warehouses. EMR reflect the temporal nature of patient care, and previous studies [Perer et al., 2015] have shown that a patient sequence of symptoms and diagnoses often correlates with their medications and procedures.

EMR describe the execution of a set of therapy and treatment activities that represent the steps required to reach a specific objective regarding some disease. These sets are called Clinical Pathway (CP) and are considered as one of best tools to increase the quality of care services [Huang et al., 2016]. As a source of data, EMR can also be subjected to exploratory analysis to infer new clinical knowledge (e.g. optimize disease progression pathways, identification of temporal patterns, etc.) and help guide decision support systems. Traditional mining techniques (e.g., clustering, statistic, nearest neighbours), despite successful in discovering latent treatment knowledge, are not suitable to analyze CP due to their additional complexity, and their dynamic and ad-hoc nature [Huang et al., 2016].

Another interesting problem to consider relates to the stepwise pharmacological therapy (i.e., choosing the most cost-effective drug therapy) for management of diseases associated with progressive conditions. While many techniques using association rules mining and frequent item-set based techniques have successfully showed relations between medications [Wright et al., 2015], they do not consider the temporal aspect, thus limiting the utility of the results.

3.2.1 Care Pathway Explorer

To extract common sequences of medication (e.g., diagnoses, treatments and the correlation between them in the patient outcome) from historical EMR, [Perer et al., 2015] proposed an hierarchical system called Care Pathway Explorer (CPE) to mine and visualize frequent event sequences from historical EMR. This system provided an interactive visualization for the frequent pathway events, leveraging on event hierarchies to present data in a user-centric way so they can be used in real world settings. CPE is able to handle concurrency, multiple levels of detail, temporal context, and patient outcomes (e.g., dead or alive). It is composed of 4 major components:

1. **Event Base**: Using Extract, Transform and Load tools, EMR data is extracted into a 4-column table (patientID, dayID, eventID, eventValue). This procedure enables the integration of different hierarchical classes of events (e.g. diagnosis and medication) in one database.

2. **Data Preprocessor**: One issue that affects the efficiency of temporal pattern mining is when we have many events occurring simultaneously (e.g., in a medical appointment the patient is required to measure weight, height, blood pressure). This creates an explosion of possibilities that hinder

frequent pattern mining algorithms. To alleviate this, we first find Clinical Event Package (CEP) that are frequent subsets of Same Day Concurrent Event (SDCE). Treating each SDCE in every patient trace as a transaction, the problem of finding CEP translates to frequent item-set mining and each CEP can be used as a super event. Each SDCE is then broken down, with basis on two-way sorting approach, as a combination of regular and super events, thus reducing the number of events in each SDCE. The result is then fed to Step 3.

3. **Frequent Pattern Analytics**: To efficiently mine temporal patterns from long sequences, an extended Sequential Pattern Mining with Bitmap Representation method is used. With this technique, event sequences are represented as binary codes, and finding frequent patterns involves only binary AND/OR operations to process those sequences. This extensions enables the method to: (1) relate to searching patterns restricted in some time interval, and (2) supporting outcomes analysis. The discovered patterns are then checked against outcomes to understand correlations.

4. **Visual Analytics**: A user interface provides two types of visualizations:

   - Overview visualization, similar to a bubble chart displaying the most frequent patterns that were mined. Each bubble represents a medical event, its size represents the number of times it occured in the discovered patterns, its color shows if the event had a positive (blue) or negative (red) outcome, and the position in the axis represents possible existing clusters.
   - Flow visualization, based on a Sankey diagram [1], where the magnitude of the connections between bubbles is represented.

Tests showed that while CPE leads to insights, it still has limitations regarding the efficiency of the mining algorithm when applied to large datasets with many concurrent events. Also, since the analytics engine requires manual specification of parameters, any changes to these parameters results in an increased waiting time.

3.2.2 Diagnosis Treatment Model

[Huang et al., 2016] proposed a generative statistical model, based on Dirichlet allocation, that associates treatment patterns with both treatment events and diagnosis labels. This model, named the Diagnosis Treatment Model, links clinician diagnosis labels and consequent treatment events together, in order to discover latent associations between diagnosis labels and treatments, and determine the contribution of comorbidities on treatment adoptions in CP.

For this model we assume that a clinical event log L consists of $|L|$ patient traces σ, each of them associated to a set of diagnosis labels $C_\sigma \subseteq C$ and a finite non-empty set of treatment events

[1] https://en.wikipedia.org/wiki/Sankey_diagram

$\{e_1, e_2, ..., e_{|\sigma|}\}$ executed in a particular patient during his stay in the hospital. A treatment event is displayed as $e_i = (a_i, t_i)$ where a is the activity type of e_i and t_i is the occurring time of e_i. The treatment activity type and occurring time stamp of some treatment event e_i is displayed as $e_i.a$ and $e_i.t$, respectively.

The algorithm starts by selecting a diagnosis label c, randomly at uniform, from the set of diagnosis labels associated to patient trace σ. Then, the multinomial distribution θ_c from the Dirichlet distribution α (Dirichlet prior of θ) is chosen, determining the treatments most likely to be assigned to diagnosis label c. This is done for every diagnosis label. Next, from the multinomial distribution associated to diagnosis label c, represented as θ_c, one treatment pattern z is sampled for each treatment event e_i in σ. Finally, a treatment activity type $e_{\sigma,i}.a$ is estimated based on the multinomial ϕ_z. In a similar way, the occurring time stamp $e_{\sigma,i}.t$ is estimated using $\varphi_{z,a}$ (generated from Dirichlet prior of φ from each treatment pattern z and treatment activity a). The number of instances of treatment pattern z assigned to diagnosis label c is denoted by $\eta_{c,z}$.

Considering the difficulty of directly sampling the parameters from a multivariate probability distribution, a Gibbs sampler [2] was selected. Assuming the treatment patterns are available, we can estimate the probability of pattern z conditioned to diagnosis label c:

$$\theta_{c,z} = \frac{\eta_{c,z} + \alpha}{\sum_{z' \in Z} \eta_{c,z'} + |Z|\alpha} \tag{3.8}$$

The probability of activity type a conditioned on treatment pattern z is given by:

$$\phi_{z,a} = \frac{\eta_{z,a} + \beta}{\sum_{a' \in A} \eta_{z,a'} + |A|\beta} \tag{3.9}$$

The probability of occurring time stamp t conditioned on treatment pattern z and clinical activity type a is, in turn, given by:

$$\varphi_{z,a,t} = \frac{\eta_{z,a,t} + \gamma}{\sum_{t' \in T} \eta_{z,a,t'} + |T|\gamma} \tag{3.10}$$

Regarding complexity, Gibbs is an efficient technique for large data sets with complexity of $\mathcal{O}(Z.A.T.C_{\max})$, where Z is the number of treatment patterns, A the number of activity types, T the number of occurring time stamps and $C_{\max}$ is the maximum number of diagnosis labels that can be associated with a single patient case. Since $A.T$ is linear, Gibbs sampler can run efficiently on large datasets.

The resulting patterns helped users to reason about the influence of comorbidities on therapy and treatment of patients in CP. In particular, the authors were able to use discovered treatment patterns to redesign CP with specific comorbidities into consideration.

[2] https://en.wikipedia.org/wiki/Gibbs_sampling

3.2.3 MIxCO

In comorbidity studies, clinical narratives were often ignored, despite having great research potential [Boytcheva et al., 2017]. Since the information contained in them can be used to increase rule effectiveness, [Boytcheva et al., 2017] proposed a cascade data mining approach for frequent pattern mining enriched with context information, including a Maximal Frequent Itemset (MFI) algorithm to mine comorbidities named MIxCO. This process begins by converting clinical narratives (i.e., patient stories written by a clinician that describes a specific clinical event or situation) from unstructured (e.g., free narratives) to structured data (e.g., EMR) using text mining techniques, which are then stored in a database. Next, MIxCO is applied to extract MFI.

Let D be a database and D^V a vertical database (i.e., where the layout is column-by-column). Let also $\mathcal{F}$ denote the set of frequent item-sets, and $\mathcal{M}$ be the set MFI, initially empty. Assume also an item $w \in C$, where C is a set of items and w is considered weak if there is no item in C with support lower than the frequency of w. P_w is the projection of the database D^V by item-sets intersection and P_w by item-sets difference. A frequent item-set $\mathcal{M}$ (i.e., item-set with a support above min-support) is considered maximal if $\mathcal{M} = \{X | X \in \mathcal{F}$ and $\nexists Y \in \mathcal{F}$, such that $X \subset Y\}$ (i.e., if $\mathcal{M}$ has no frequent super-sets).

MIxCO starts by generating the database $D^{V'}$ from D, by deleting tuples of items that fail to meet min-support. We now proceed to obtain MFI by the weakest item $w \in C$ in $D^{V'}$. Since we do not know if MFI contains w or not, we divide $D^{V'}$ in two subsets using projections: $P_W(D^{V'})$ and $\overline{P_W(D^{V'})}$. The algorithm is recursively applied to search for all MFI in $P_W(D^{V'})$. Let the result set of all MFI in $P_W(D^{V'})$ be $\mathcal{M}_w$. We then add to each of the items in $\mathcal{M}_w$ the item w, and obtain the MFI that contain w:

$$\mathcal{M}_w^{'} = \{Y | X \in \mathcal{M}_w, Y = X \cup \{w\}\} \tag{3.11}$$

Let β be the set of items of $\overline{P_W(D^V)}$ that were reduced due to low frequency. These items cannot be reduced any further since they have low frequency in combination with w, but since they pass the min-support test in $D^{V'}$ they can be items of MFI that do not contain w. $\overline{P_W(D^{V'})}$ is updated with those item-sets that contain members of β:

$$\mathrm{P}_w(D^{V'})^U = P_w(D^{V'}) \cup \{\langle c, y \cup (z \cap p(x))\rangle \mid \langle x, p(x)\rangle \in \beta, \langle c, z\rangle \in P_W(D^{V'}), \langle c, y\rangle \in P_W(D^{V'}) \tag{3.12}$$

Let $\mathcal{M}_w$ be the result of applying recursively the algorithm to search for all MFI in $P_w(D^{V'})^U$. Then, the $\mathcal{M}$ of D is produced as follows, and the last step is to remove all existing frequent patterns that are not maximal $\mathcal{M}$.

$$\mathcal{M} = \mathcal{M}_w^{'} \cup \mathcal{M}_w \tag{3.13}$$

Finally, data is enriched with context information by adding semantic attributes to each event. Let $A = \{a_1, a_2, ..., a_k\}$ be the set of interesting attributes. The context Q for some patient $p_i \in P$ is defined as the set of attribute-value pairs from event profile information:

$$Q(p_i) = \{\langle a_1, q_1\rangle \ , \langle a_2, q_2\rangle \ , ..., \langle a_k, q_k\rangle\} \tag{3.14}$$

From $Q(p_i)$ we generate a feature vector $v(p_i) = (v_{1i}, v_{2i}, ..., v_{mi})$ where each attribute $a_j \in A$ with N_j possible values is represented by N_j consecutive positions in the vector.

Some types of comorbidities, in particular those related to mental health, do not imply multiple diseases, but are often the result of an inability to make a single diagnosis accounting for all symptoms. The experiments showed MIxCO was capable of solving these issues with the longest MFI, and the method was also proved efficient at processing big dense datasets in relatively small min-support values. However, noise was not taken into account when testing. More interesting were the discovered correlations between phenotypes and comorbidities after adding context information.

3.2.4 Prediction using Sequential Pattern Mining

[Wright et al., 2015] studied the feasibility of identifying temporal relationships between medications with SPM. In particular, the authors wanted to check if SPM is useful in predicting changes to patient medication regimes, and to assess how useful is the patient history of prior medication changes when making a prediction.

First, a decision on the desired level of granularity (e.g., medication name, generic medication name only, generic medication and dose) to mine the data is made so the number of generated patterns can be reduced. Then, a mapping table attributing a class to every drug is loaded into the database to be mined. Leveraging on this table, the data from pre-existing database is transformed into a horizontal transaction format as required by the Constraint Sequential Pattern Discovery using Equivalence classes algorithm . This algorithm mines frequent sequential patterns using temporal joins along with efficient lattice search techniques and provides for timing constraints, enabling fast discovery of sequential patterns by avoiding multiple database scans. Making use of the package arulesSequence [3] in R programming language for statistical computing, frequent patterns were mined with a defined support. The top sequential patterns were then chosen by grouping them by length and by support.

For rule prediction, each mined pattern was transformed into an antecedent-consequent pair (i.e. a rule), where given the antecedent the rule predicts the consequent. Each patient sequence is then transformed into a base stem and the associated next drug. The base stem shows the history of patients medication changes prior to their most recent prescription for a given illness. The prediction of the

[3] http://cran.r-project.org/web/packages/arulesSequences/index.html

patient's next drug is made by looking at rules whose antecedent matched the base stem, afterwards ranking them by support. The base stem length can be adjusted if the desired number of predictions cannot be achieved. Testing results, using 5 attempts, showed that a prediction of 97.7% was achieve for the next class of medication, while on the generic medicine level the accuracy prediction was 71.9%. In the medicine class level, while the results using 1-length stems were always better than 0-length ones, 2-length stems only achieved better predictions for patients with 2 drug classes. Even more interesting, using 3, 4 or 5-length stems to make predictions was always worse or equal than using 1 or 2-length stems. Similarly, in the generic drug level the best predictions were made using only base stems of length 1 or 2.

Regarding the importance of patient history of prior medication changes when making a prediction, the base stems were reduced to one item at a time from the beginning of the stem. In tests assuming 3 prediction attempts on the drug class level, using 10-fold cross validation, the average number of patients for whom the next drug could be correctly predicted was 90.0%, while on generic drug level this metric was down to 64.9%

To better assist clinicians with the prediction, a directed graph (i.e., edges have a direction associated) visualization name GraphViz[4] was used. This type of graph shows interesting visual representations of stepwise medical therapy. Possible uses include (1) the planning of a patients treatment regimen, (2) helping patients to understand the typical progression of medications used to treat their condition, and (3) epidemiological uses for those who wish to monitor and study the behavior of prescribers.

Overall, results showed that performance was always better at the class level than at the drug level, due to number of drugs being greater than the number of classes (38 vs 10). Using 1 or 2-length base stems produces higher precision on predictions, and using the entire stem was not always better. The results support the idea that SPM is useful to make suggestions to what drug to prescribe next, and accurate predictions could be made without using all the patient medication history.

3.3 Summary

This chapter presented different data mining techniques to extract relevant patterns in ordered sequences of events. Past attempts to mine databases of medical events were also reviewed. Table 3.1 presents an overview of the surveyed approaches. The column efficiency features NA when efficiency assessment was not available, the symbol $-$ when the technique is not efficient, $\pm$ means that the technique is efficient only in certain conditions, and $+$ denotes efficient techniques. The column type indicates what is the main concept behind each study and the medical column denotes if the study was performed in a medical database or not.

[4] http://www.graphviz.org/

Table 3.1: Overview on the data mining techniques.

Study	Type	Efficiency	Medical
[Lu et al., 2008, Lu et al., 2010]	Structural Relation Patterns	+	N
[Tax et al., 2016]	Local Processes	−	N
[Leemans and van der Aalst, 2015]	Local Processes	+	N
[Mannhardt et al., 2018]	Event Abstraction	−	N
[Chapela-Campa et al., 2017]	Frequent Patterns	+	N
[Perer et al., 2015]	Frequent Sequences	±	Y
[Huang et al., 2016]	Statistical Model	NA	Y
[Boytcheva et al., 2017]	Frequent Item-sets	±	Y
[Wright et al., 2015]	Sequential Patterns	NA	Y

The algorithm developed by [Lu et al., 2008, Lu et al., 2010] proved to be an efficient method to extract structural relation patterns, i.e. concurrent,exclusive and iterative patterns, however the structural relation patterns were found prone to over-fitting problems. An algorithm, based on local process models, developed by [Tax et al., 2016] solved that problem by returning a set of resulting patterns, however it was found not efficient. Also based on local process models, [Leemans and van der Aalst, 2015] developed an efficient algorithm when setting optimal threshold values. [Mannhardt et al., 2018] algorithm, although presented many features like the ability to deal with noisy data, reoccurring and concurrent behavior, and shared functionality proved to be inefficient in tests. While all the previous algorithms generated the patterns, the algorithm developed by [Chapela-Campa et al., 2017] receives a model that focuses on finding patterns in that model. This technique is not only efficient but it also always returns the correct frequent patterns. Regarding techniques applied to medical databases, the algorithm developed by [Perer et al., 2015], although generated insights, has limitations regarding efficiency. Techniques developed by [Huang et al., 2016, Wright et al., 2015] also generated insights on treatment patterns and prediction of medications, however they were not tested to assess their efficiency. The algorithm developed by [Boytcheva et al., 2017] proved to be an efficient way to discover correlations between phenotypes and comorbidities, despite the fact that noise in the data was not taken into account when tested. Given the context of this work, Cohort Studies and Frameworks based around the concept of ”change” and ”expertise” are also considered.

4. Methods

Based on the KDD research model described in Section 1.2, this section presents the methods used to discover interesting patterns in medical prescription databases using data mining techniques. In Section 4.1 we describe in detail the selection of the data set and the pre-processing steps necessary to make it suitable for the data mining algorithms. Then, in Section 4.2 we explain the data transformation and the application of the data mining algorithms. Finally, in Section 4.3, we show how the generated patterns were evaluated.

4.1 Data Selection and Pre-Processing

Mining interesting patterns requires first transforming a database containing a chronologically ordered sequence of prescriptions into one that indicates how patients are being medicated throughout time, e.g. in terms of the different active substances and patient treatment regimes.

In an electronic prescription databases one is expected to find fields regarding patients, including a unique identifier for the patient, as well as descriptive attributes like age, gender, and location (e.g., administrative district). These last fields, in particular, must be carefully examined and consolidated, since in the context of longitudinal studies patients can change throughout time. Expected fields regarding active substances include a unique name for the substance, the type of administration, prescription dosage, package size, and number of packages. Finally, attributes for characterizing the context of a prescription include a date, an identifier for the doctor making the prescription, and the doctor's medical speciality.

None of the fields above contains an indication on how many days a given prescription can maintain a person under treatment. To estimate this, we can leverage the concept of Defined Daily Dosage (DDD), which corresponds to the assumed average maintenance dose per day for a medicine, considering its main indication in adults. Although the DDD does not necessarily reflect the prescribed daily dosage, it gives a rough estimate of consumption, providing a fixed unit of measurement independent of price, currencies, package size, and strength. Considering the DDD, the duration, in days, of a treatment can be estimated by:

$$\text{Nr Days} = \frac{\text{Nr Packages} \times \text{Nr Pills per Package} \times \text{Dosage}}{\text{DDD}} \tag{4.1}$$

This information can be used to reconstruct the time intervals for which each patient was subjected to a certain treatment, using the prescription date as the start of the treatment. Depending on the associ-

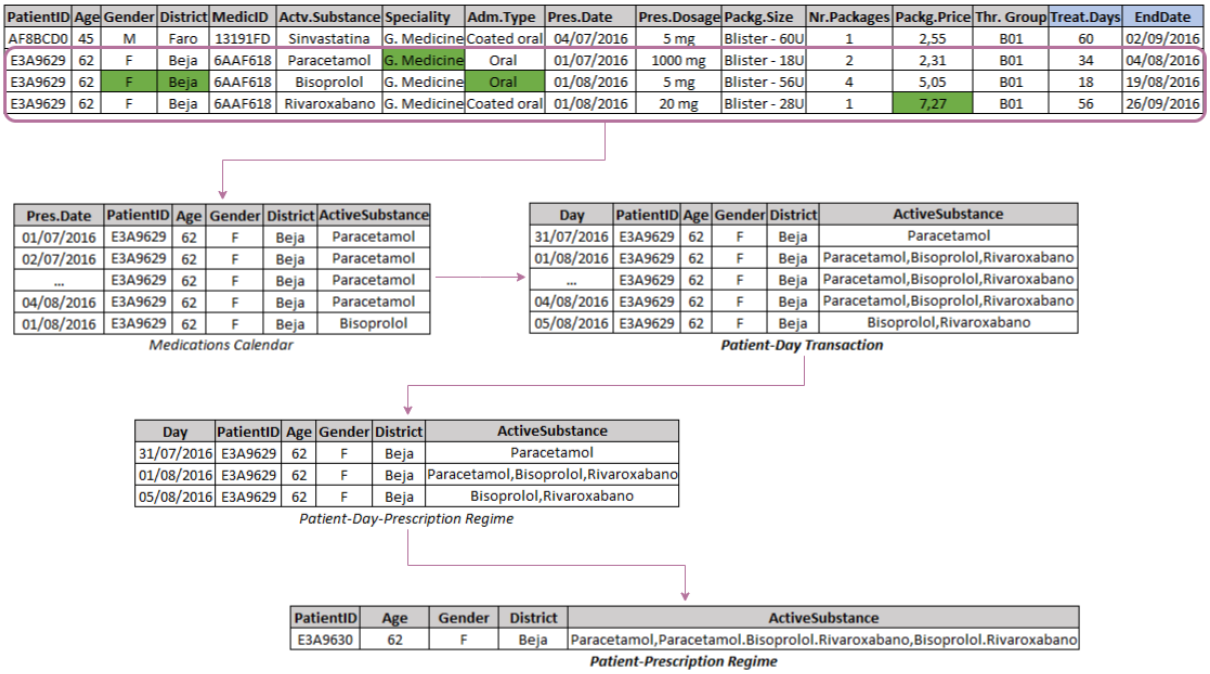


Figure 4.1: Example illustrating the sequence of transformations involved in data pre-processing, before the application of data mining algorithms.

ated therapeutic group of a given prescription, an active substance can have multiple DDD's. In the case of our study, to determine the correct DDD of a given prescription, we made a manual mapping between the active substance, the therapeutic group, and the medical speciality, using information collected from the World Health Organization[1] website, combined with doctor Paulo Nicola medical expertise. Using this information, the original records were augmented with attributes for the therapeutic group, number of days associated to the treatment, and the expected end day for the treatment.

Patient-Day transactions are formed by grouping records by patient and day. Each of these transactions correspond to all the active substances prescribed to a particular patient on a given day, according to the estimates provided by Equation 4.1. In the example shown in Figure 4.1, we can see that the patient identified by E3A9629 took Paracetamol continuously from 01/07/2016 until 04/08/2016. He also started taking Bisoprolol and Rivaroxabano from 01/08/2016. The corresponding Patient-Day transactions for this example shows that on 31/07/2016 this patient was only taking Paracetamol, but on the next day he was taking Paracetamol, Bisoprolol and Rivaroxabano at the same time. In cases where a patient has multiple prescriptions of the same medication with overlapping dates between them, can be combined into a single treatment period by increasing the end date of the first prescription with the treatment days of the subsequent overlapping prescriptions.

Since transactions can contain fields with missing or contradictory values (e.g. multiple gender

[1] http://www.whocc.no/ddd/

changes, missing age, missing speciality), I attributed age, gender, and district to the most frequent values inside the transaction (in cases where no frequent value was available inside transaction, it was used the most frequent value associated with the patient).

To identify patient treatment regimes, consecutive records of the same active substances belonging to the same patient from Patient-Day transactions are truncated to only one record (most recent date) meaning one treatment. Now, Patient-Prescription Regime transactions are formed by grouping all the previous records by patient id. Each transaction corresponds to a period of continuity in terms of a patient's treatment regime. These transactions enable the assessment on treatment changes as an indicator of a patient adherence to a different treatment regime, including possible causes. To better identify the treatment regimes, I assumed that two regimes with equal active substances constitute one treatment if the time interval between them is less than 2 days, otherwise they are treated as two separate regimes. In the example shown in Figure 4.1, consecutive transactions with three substances (i.e., of Paracetamol, Bisoprolol and Rivaroxabano) were grouped into a single transaction Patient-Day-Prescription Regime representing changes in treatments.

To identify the sequential patterns, the input consisted of the prescriptions from each patient, chronologically ordered. In an optimized version, the entry sequences contained all the prescriptions with less than 30 days between them, i.e patients with very spaced prescriptions are represented by several sequences, instead of only one.

For both association rule and frequent item-set mining algorithms, different sub data-sets were created to facilitate prescription comparisons between age groups, gender and districts.

4.2 The Data Mining Process

The process to discover AR began with the Patient-Prescription Regime dataset. In this dataset each record associated one patient to all the prescribed medications. The Apriori algorithm begins by determining the support of each 1-itemset from the dataset. Those item-sets that fail to meet the user-defined minimum support, will not be used to generate new item-sets (e.g. if Bisoprolol does not meet the minimum support, then any subsequent combination containing Bisoprolol cannot be frequent according to the Apriori principle). The remaining ones, i.e. the frequent item-sets, are used to generate k-item-sets using the frequent k-1 item-sets from the previous step. This process continues until no new frequent item-sets are generated.

The process to discover sequential patterns begins with the Patient-Day Transactions dataset. Each record in this dataset corresponds to all the medications prescribed taken by a patient on a specific day. Before applying the PrefixSpan algorithm, an extra step must be applied to the records in the dataset. Each record containing two or more medications are expanded into N new records, where N is the number of different combinations possible, as shown in Figure 4.2.

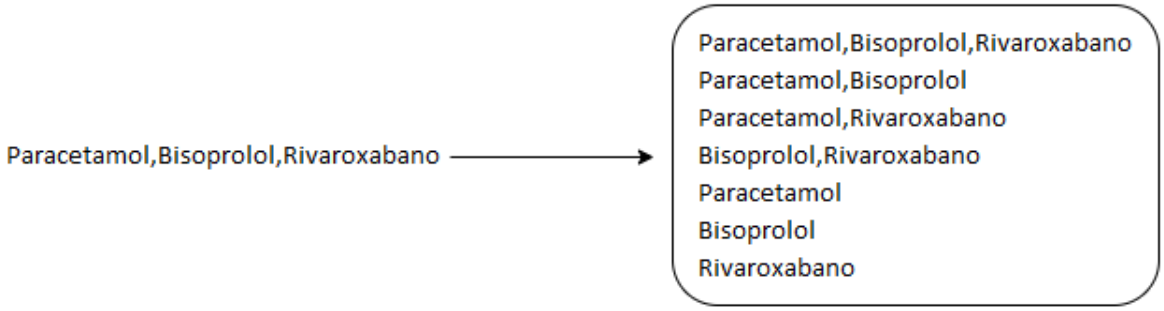


Figure 4.2: Example illustrating the transaction expansion of a 3 item sequence.

After this data transformation step, the resulting dataset is then fed to the PrefixSpan algorithm. The algorithm starts with an initially empty prefix and appends each item. The support of the different items (medications) are calculated, and those that fail to meet the minimum support threshold are eliminated. The remaining ones are appended with the remaining frequent items and the process continues until no more item-sets are available to be appended.

4.3 Evaluation Methodology

To evaluate of the generated rules and sequences, we based our evaluation process in the framework Rule Changing + Relevance Feedback described in subsection 2.3.5.1. This framework is based on the notion of change to identify interesting patterns, i.e, stable patterns are overlooked since they are already known and don't provide any additional knowledge, while the changing ones are object of a more thorough analysis. To help make this analysis we used a combination of objective and subjective measures, complemented with the technical expertise of doctor Paulo Nicola.

To evaluate the AR generated, we used the objective metrics Lift, Confidence and Support in this order to make an initial ordering, since Support has little to no variation in our experiments. In medical context, the interest of AR are typically correlates to their causality, so we adopted the Retrospective Cohort Study process to identify causal rules. In this process for each of the initial top 10 AR, we start by randomly selecting a pair of matched records, i.e. one record contains the antecedent (LHS) and the other does not, while the consequent (RHS) is blinded. This matched pair was then removed from the dataset, and the process is repeated until no more pairs can be found. The Odds Ratio of the AR is then calculated as the ratio between the records whose exposure group contains consequent and the non exposure group does not, and number of records whose exposure group does not contain response variable and non exposure group contains response variable. Since the selection of the matched pairs is random (Odds Ratio of a AR varies each time), we repeated the process 5 times, and attributed the

average value of the calculated Odds Ratio to the corresponding rule.

The process to evaluate the generated sequential patterns was based on the length (i.e., with a length greater than 2) and the relative frequency of the patterns. On both AR and sequential patterns, the most promising results were manually examined by doctor Paulo Nicola to help identify new trends in medical prescriptions.

4.4 Summary

This chapter presented a detailed description on the methods used in this work, including the intermediate artifacts generated in the process. Since real world data is usually incomplete (e.g. missing fields); is noisy (e.g. outliers) or simply inconsistent (e.g. attribute names, price), a pre-processing phase was necessary to consolidate and enrich the data. This was then transformed into the required dimensionality to be used as input for Apriori and PrefixSpan algorithms. The identification of interesting patterns in the results, was based on the Rule Changing + Relevance Feedback framework. Additionally, we also used the concept of Retrospective Cohort Studies to identify PIR and used the expertise of doctor Paulo Nicola to evaluate the most promising rules discovered. In the next chapter, we present an initial characterization of the dataset, together with the results obtained from applying the chosen DM techniques.

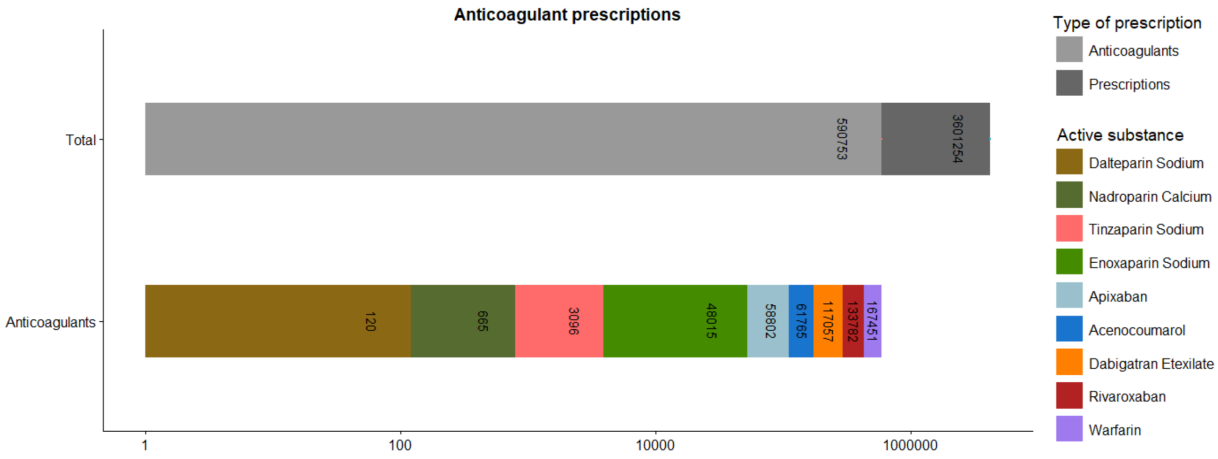


Figure 5.1: Number of prescriptions for anticoagulants.

5. Results

In this section I present the results obtained by performing the methodology described earlier. I began by making an initial description of the data-set. Then, a segmentation of the population was made, and multiple subsets were created, based on the intial description data (e.g., Age Interval, District, Year).

5.1 Dataset Characterization

An initial statistical characterization from the Portuguese medical prescriptions database regarding years 2015 and 2016 has been made, focusing on prescriptions for common anticoagulants (i.e., chemical substances used in therapy for thrombotic disorders, that prevent or reduce coagulation of blood, thus prolonging the clotting time).

Starting with general statistics, the horizontal bar plot shown in Figure 5.1 presents the total number of prescriptions in the 2 year period, comparing it with the number of prescriptions involving one of the 9 different anticoagulants considered in the study. Stacked bars are used to present the number of prescriptions associated to each of the anticoagulants. In a total of 4192007 medical prescriptions, from 30396 different medical doctors and involving 125196 different patients, approximately 14.1% correspond to anticoagulants with 28.3% of those corresponding to the prescription of Warfarin, the most common anticoagulant and also the most common medication being prescribed overall in Portugal[1].

[1] http://www.infarmed.pt/documents/15786/17838/Relatorio_Anticoagulantes_05012015.pdf/c8c00f82-0541-4bcf-9eb2-80db9f569f99

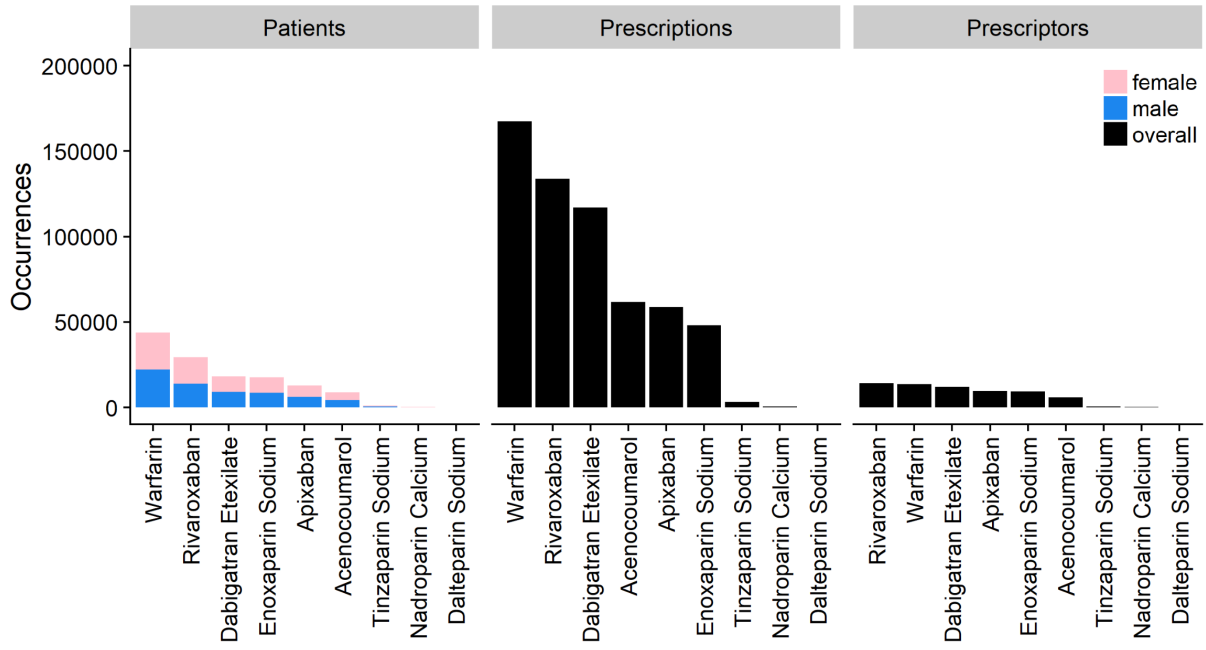


Figure 5.2: Number of patients, prescriptors and prescriptions, for different anticoagulants.

The bar plot shown in Figure 5.2 shows the number of patients with prescriptions for anticoagulants, separating the male and female patients, together with the number of different medical doctors performing these prescriptions, and the total number of prescriptions. This figure further highlights the fact that Warfarin is the most commonly prescribed anticoagulant, with Rivaroxaban (i.e., the most popular NOAC) in the second place. The number of medical doctors prescribing Rivaroxaban is in fact slightly larger than the number of doctors prescribing Warfarin, although the number of prescriptions for Warfarin is much larger. The plot in Figure 5.3 presents the distribution for the number of patients with anticoagulant prescriptions per age group, while the two bar plots shown in Figure 5.4 present anticoagulant prescriptions per month, both in terms of absolute numbers and in terms of a percentage of the total number of anticoagulant prescriptions. As expected, anticoagulant prescriptions are much more common in patients over 75 years of age. The monthly fluctuations shown in Figure 5.4 also reflect changes in the total number of monthly prescriptions and number of distinct doctors prescribing anticoagulants, and the same figure also shows that the relative number of prescriptions for the different types of anticoagulants remains relatively stable (although the percentage of NOACs versus traditional anticoagulants seems to be increasing).

The bar plots shown in Figure 5.5 present the 5 most common active ingredients (green bar indicates an anticoagulant) co-prescribed with each of the 2 most popular anticoagulants, considering a period of 1 month in between prescriptions, in an attempt to assess the prevalence of prescriptions that might give rise to clinically relevant drug-drug interactions in the population of patients taking anticoagulants (e.g., drugs such as Warfarin, when taken with other drugs such as NSAIDs, aspirin, or macrolide

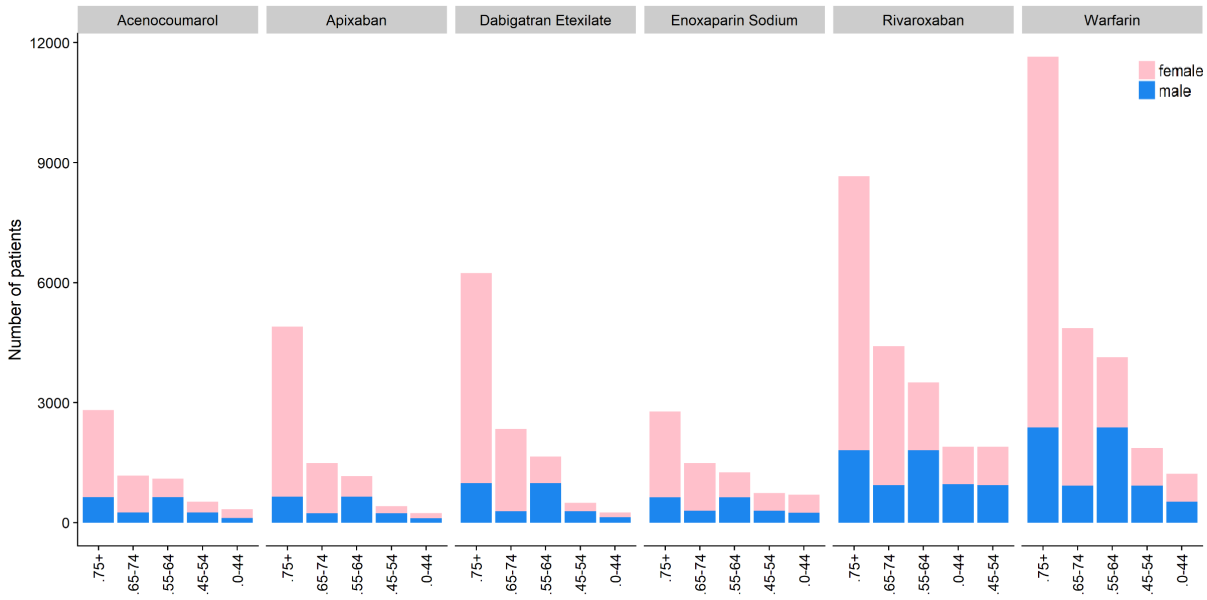


Figure 5.3: Number of patients with anticoagulant prescriptions, per age group and gender.

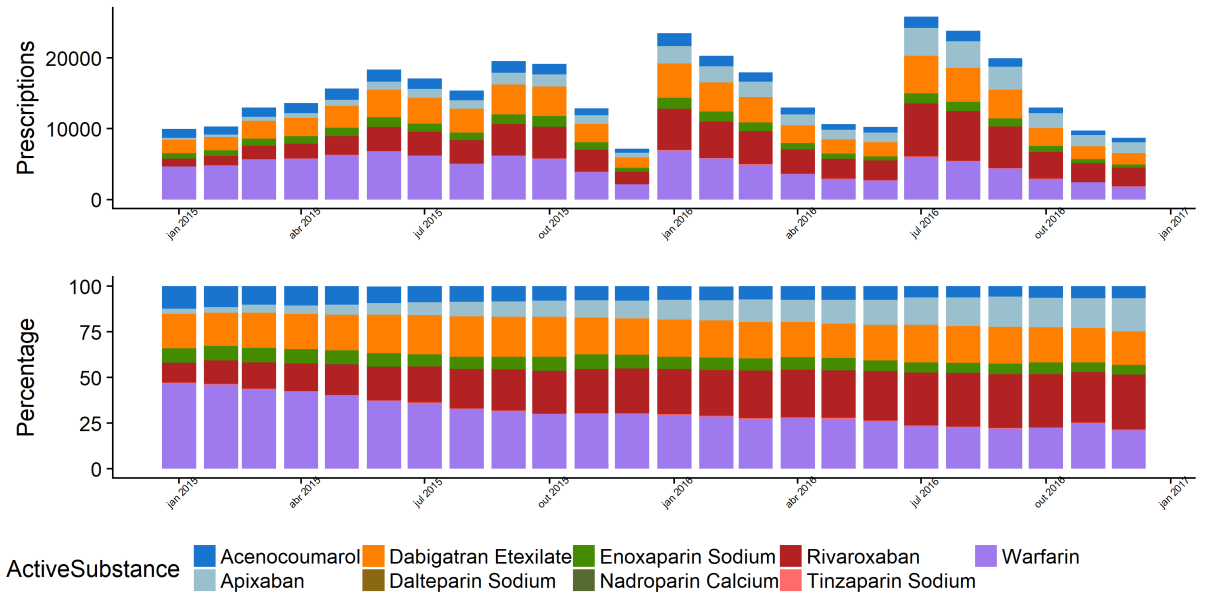


Figure 5.4: Monthly distribution for the number of prescriptions of anticoagulants.

antibacterials, gives rise to an increased bleeding risk). The results show that not many of the patients have co-prescriptions for different types of anticoagulants, perhaps due to a lack of changes in the treatment regimen over time.

The maps shown in Figure 5.6 present the spatial distribution, for the same anticoagulants from Figure 5.5, for patients with anticoagulant prescriptions, as a fraction of the total number of inhabitants per municipality collected in the 2011 national census. In turn, the maps shown in Figure 5.7 present the

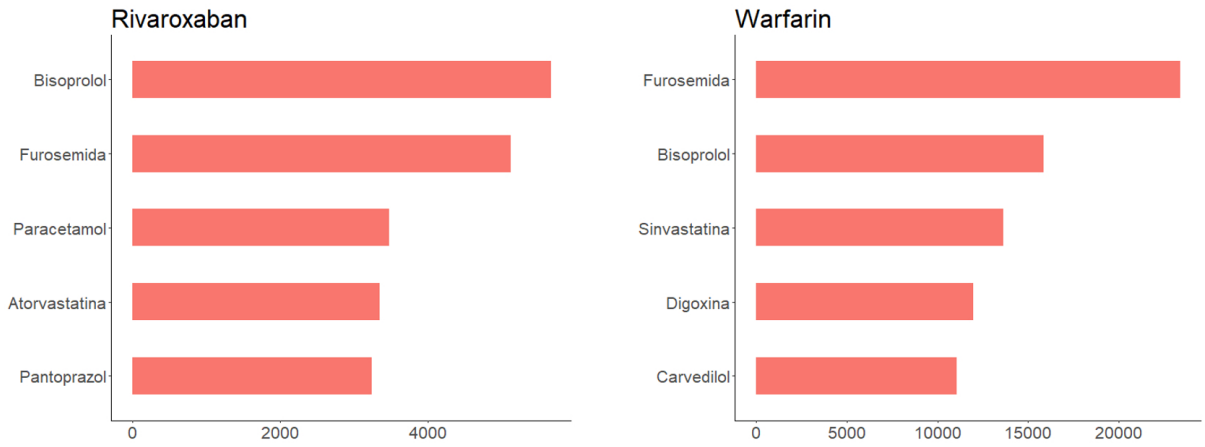


Figure 5.5: Top 5 medications prescribed together with different anticoagulants.

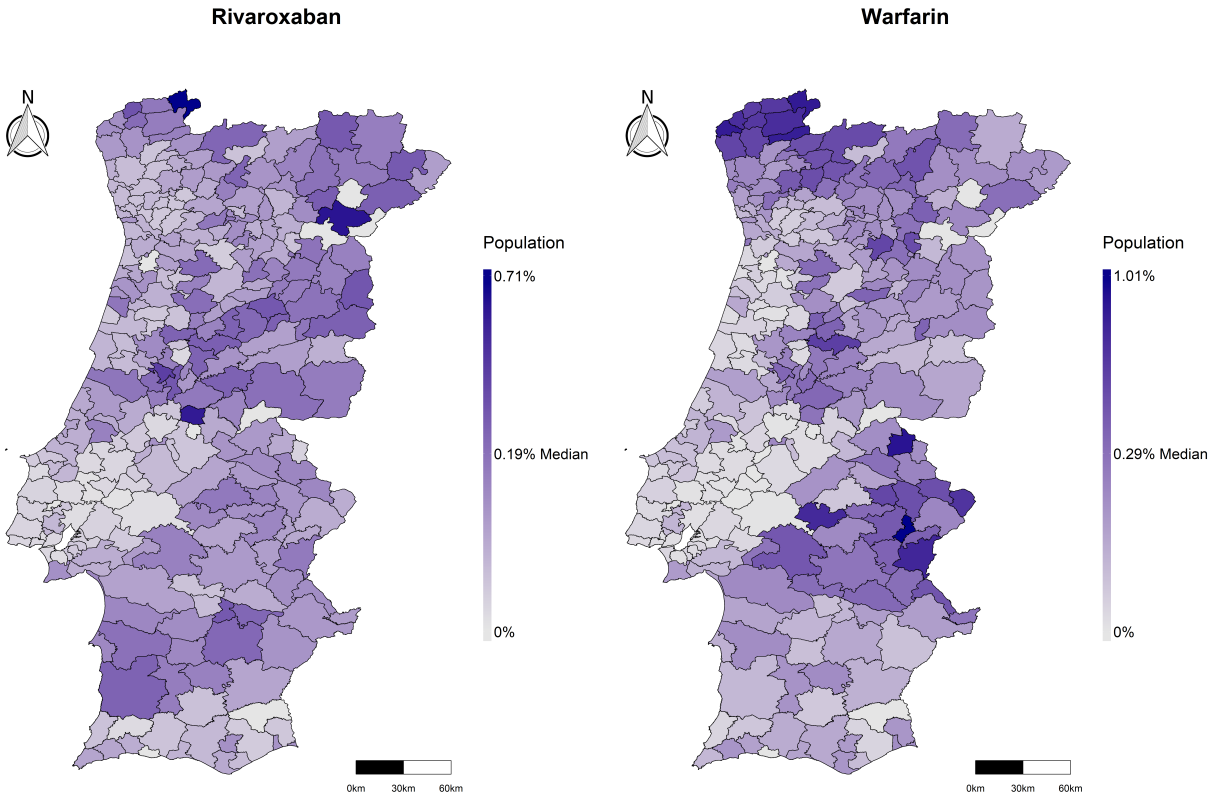


Figure 5.6: Spatial distribution of patients with prescriptions for anticoagulants.

number of medical doctors prescribing different anticoagulants per municipality, as an absolute number. The figures show a disparity in terms of the relative incidence of prescriptions for different types of anticoagulants.

Finally, Figure 5.8 presents a Sankey diagram illustrating the number of patients that changed their treatment regimen over time. The anticoagulant prescriptions for the 2 year period were split into periods

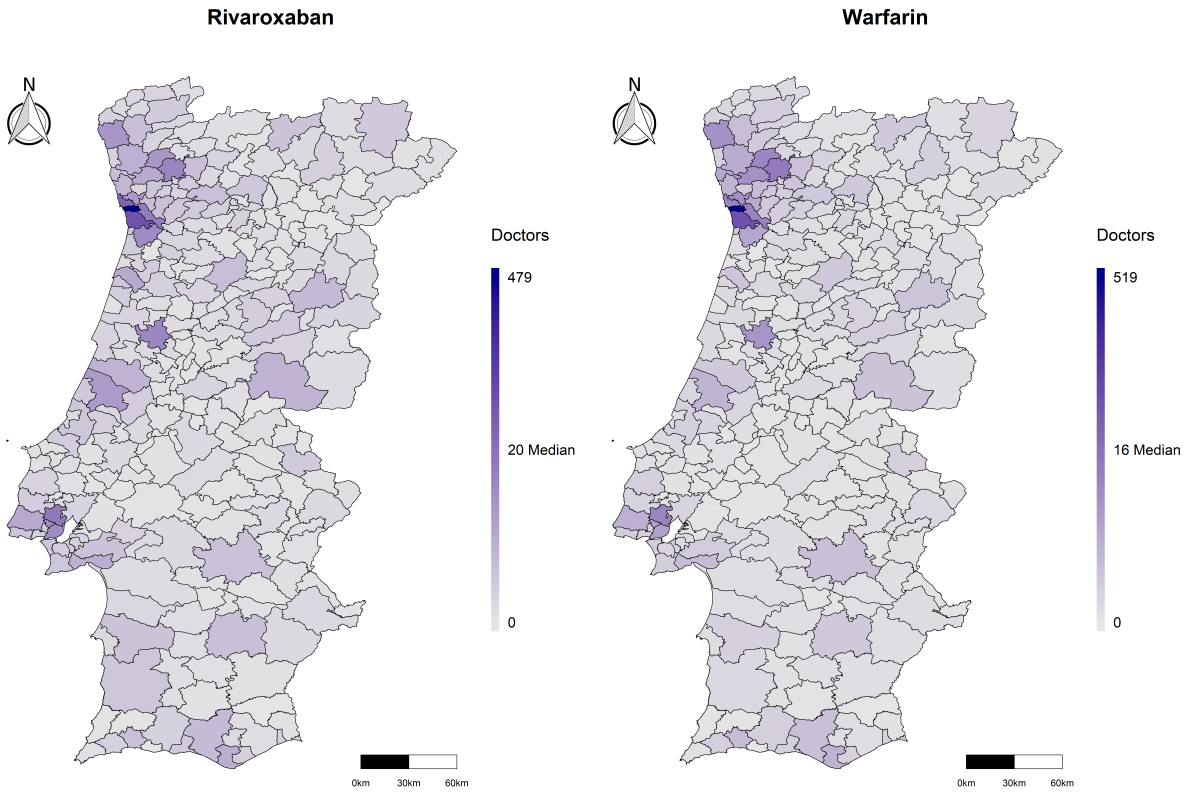


Figure 5.7: Spatial distribution of medical doctors that prescribed anticoagulants.

of 3 months, and for each period we show the number of patients that changed from one anticoagulant to another. The figure shows that NOACs are increasingly being prescribed.

5.2 Results for Association Rules

We used association rules to characterize the relationships between medications prescribed (antecedent) and the next one to be prescribed (consequent). To visualize the discovered association rules, we used nodes and arrows to connect the antecedent to the consequent of a rule. The size and color

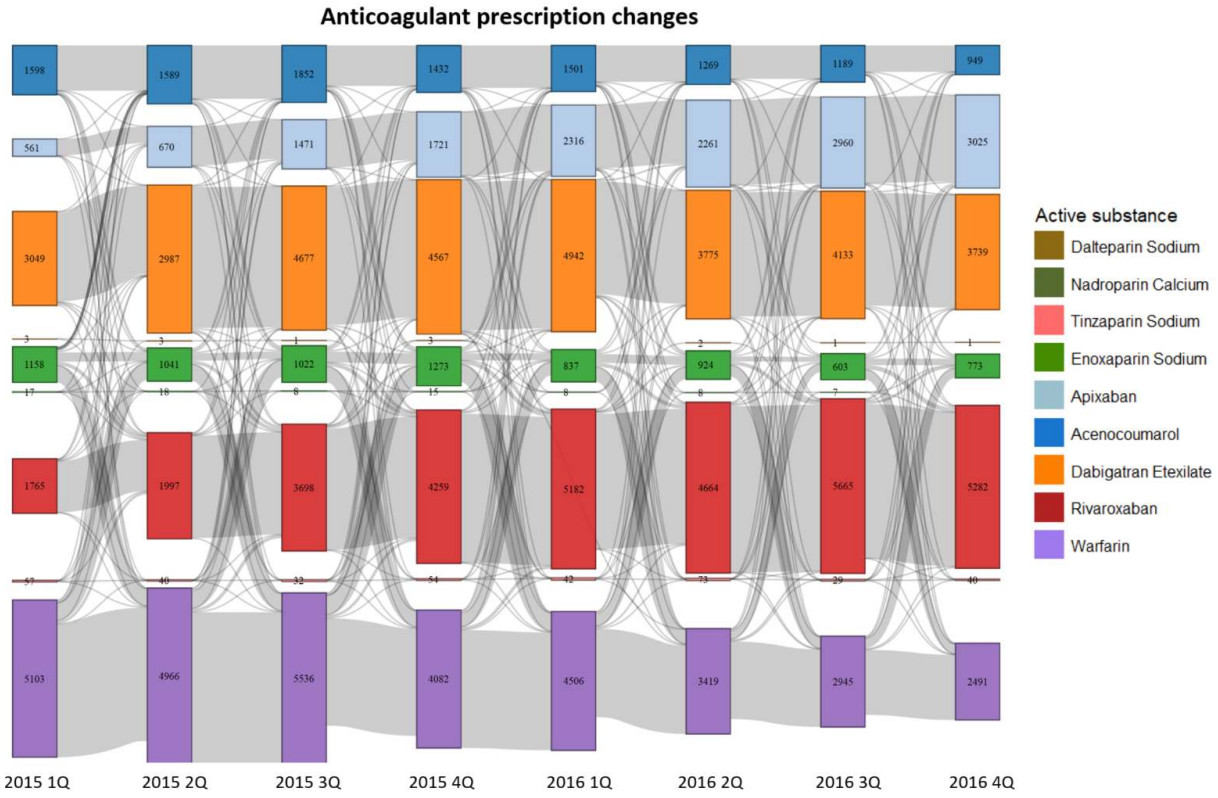
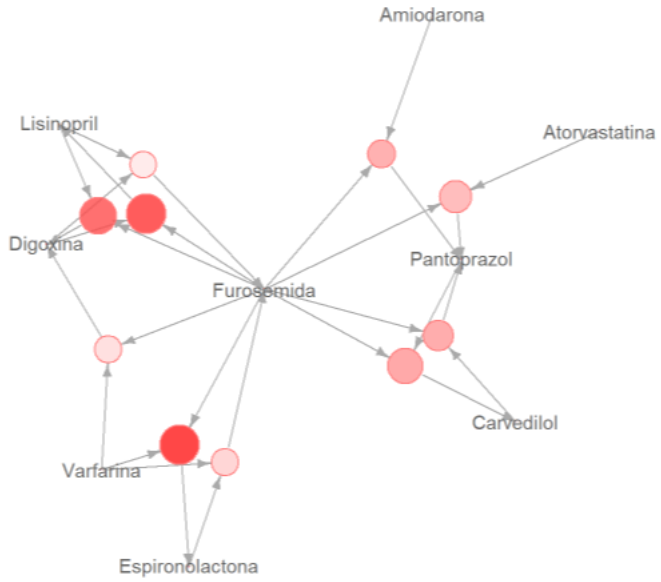


Figure 5.8: Anticoagulant prescriptions over time.

of a node provide a visual representation of the lift and odds ratio value of a rule. Rules that have large nodes represent potentially interesting rules, while those with darker colors indicate causal rules. For example, looking at figure 5.9 representing the top 10 rules (ordered by lift and odds ratio in this order) in Portugal, the rule Furosemida,Varfarina $\Rightarrow$ Espironolactona has a high lift and odds ratio, implying that Espironolactona is specifically used with Furosemida and Varfarina, and a good candidate for a causal rule. However, it is interesting to note that Varfarina,Espironolactona $\Rightarrow$ Furosemida has a smaller lift and odds ratio. This implies that Furosemida is frequently used with other medications and can be considered as a commonly used medication.

Observing the top rules by gender in Figure 5.10, it is possible to see that the rules associated with the male patients have a greater lift when compared with the female ones. Also, it is interesting to see that not only the female top rules contain 33% more medications than the male's, but also that the NOAC Rivaroxaban, already appears in in top rules.

Looking at the top rules by age groups in Figure 5.11, it is possible to see that in the age group 0-44, there are evidences that support that a causal rule between the taking of Escitalopram, Nebivolol and Candersartam, implying the taking of Warfarin, the only anticoagulant present in this group. In group 45-54 there are two anticoagulants, the older and cheaper Warfarin and the more expensive NOAC named Rivaroxaban. Also in this group, there is a possible causal rule between the taking of Nebivolol together



Antecedent	Consequent	Lift	OddsRatio
Digoxina,Furosemida	Lisinopril	14,496	10
Furosemida,Varfarina	Espironolactona	14,464	9
Furosemida,Lisinopril	Digoxina	13,531	8
Furosemida,Pantoprazol	Carvedilol	13,118	3,5
Atorvastatina,Furosemida	Pantoprazol	11,831	3
Carvedilol,Furosemida	Pantoprazol	11,355	5
Amiodarona,Furosemida	Pantoprazol	10,454	4
Espironolactona,Furosemida	Furosemida	10,231	2
Furosemida,Varfarina	Digoxina	10,06	1
Digoxina,Lisinopril	Furosemida	9,986	1

Figure 5.9: Top 10 association rules from the entire dataset.

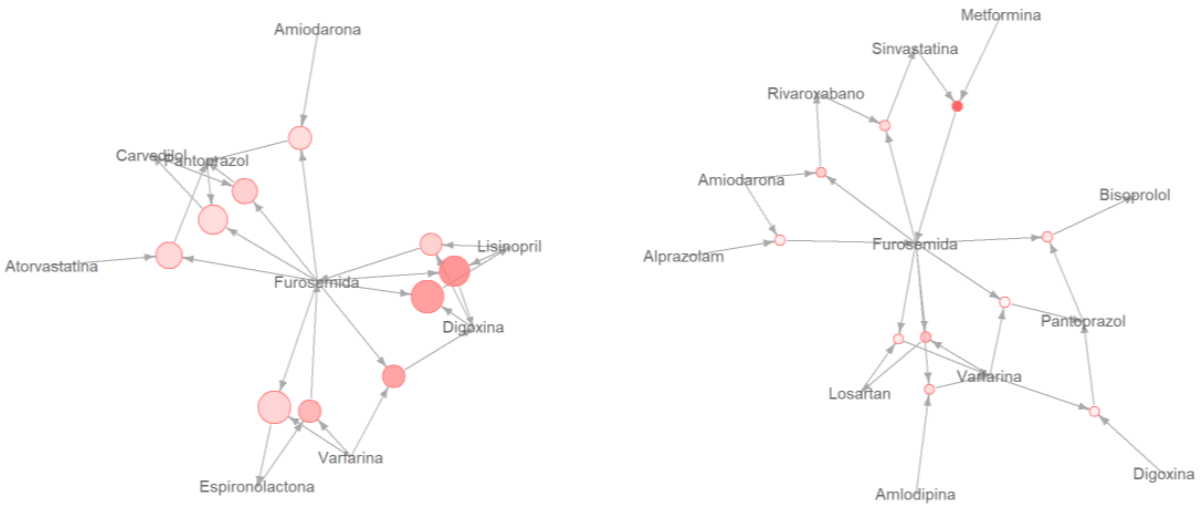
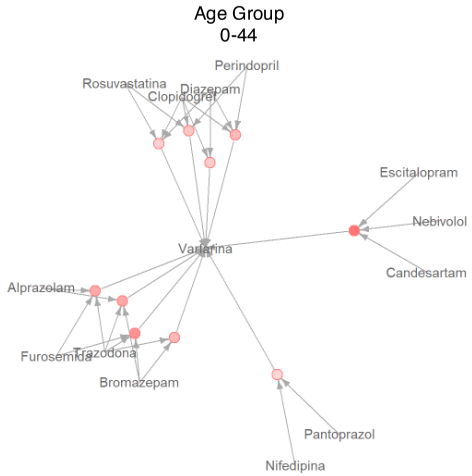
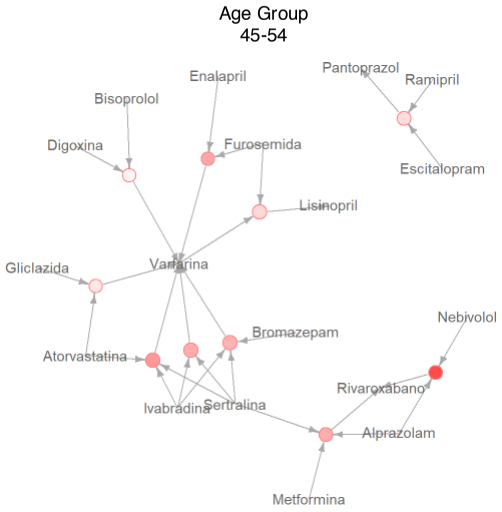
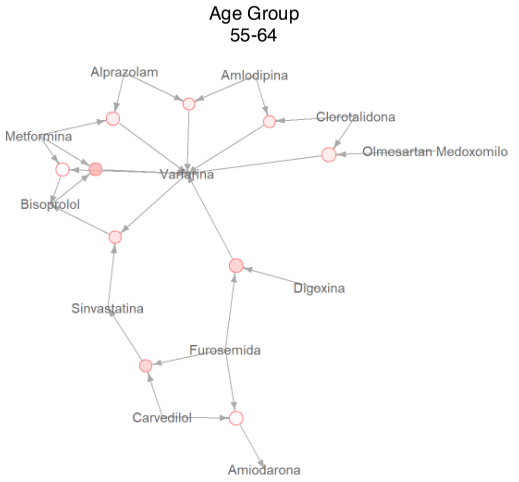


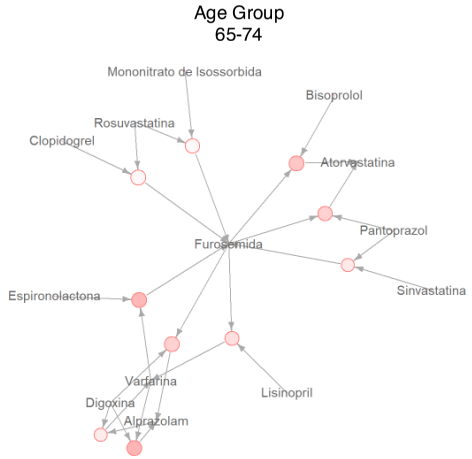
Figure 5.10: Comparison between top rules in male (left) and female (right) patients.

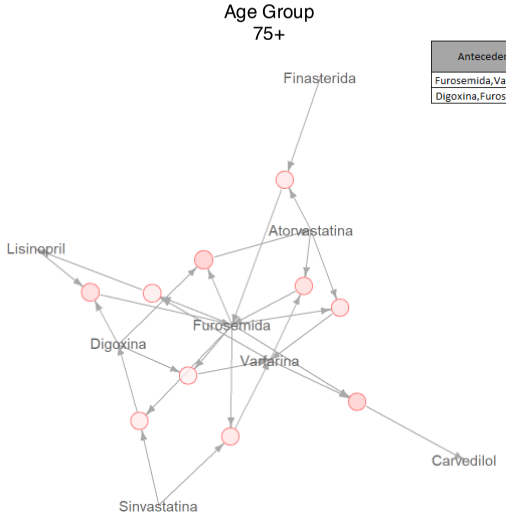
with Alprazolam and the consequent taking of Rivaroxaban.











Antecedent	Consequent	0-44		45-54		55-64		65-74		75+	
		Lift	Odds Ratio	Lift	Odds Ratio	Lift	Odds Ratio	Lift	Odds Ratio	Lift	Odds Ratio
Furosemida,Varfarina	Lisinopril	-	-	6,977	6,4	-	-	-	-	8,807	1,5
Digoxina,Furosemida	Varfarina	-	-	-	-	7,326	3,99	-	-	8,426	1,13

Figure 5.11: Top rules by age group.

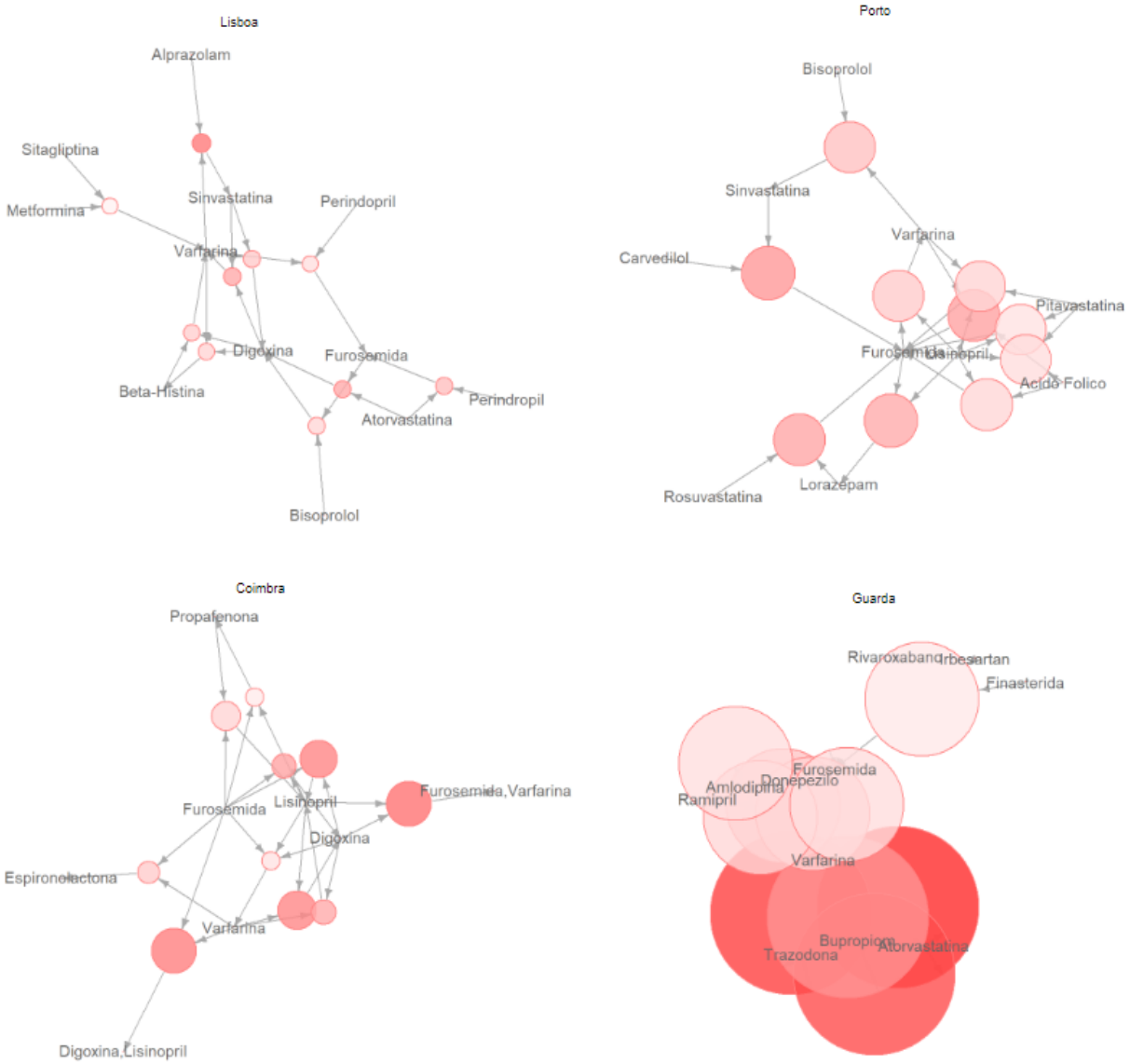


Figure 5.12: Top rules according to the different districts.

Another type of comparison was made using the districts where the prescriptions were made. Comparing the Lisbon and Porto districts (biggest population centers in Portugal with similarly sized population) in Figure 5.12, we can observe that top rules in Porto have a significantly larger lift than those in Lisboa. Also in the same figure, the top rules in the coastal district of Coimbra and the inner district of Guarda are also shown. However due to the low population in inner districts, and thus fewer prescriptions, the rules discovered in these types of districts typically have a bias associated.

5.3 Results for Sequential Patterns

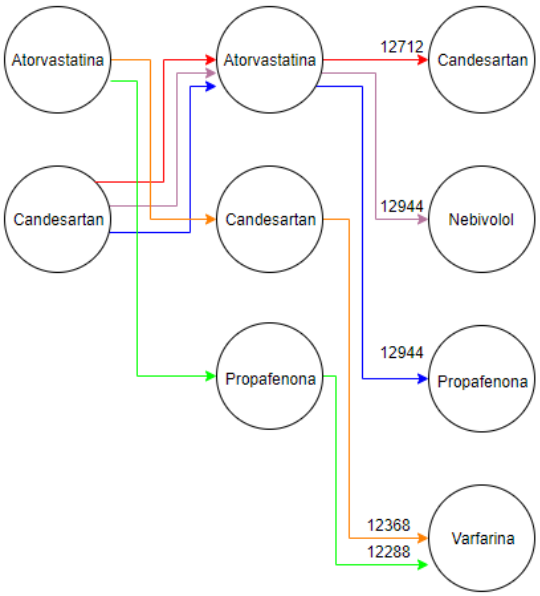
To discover statistically relevant patterns present in Patient-Day transactions, we used PrefixSpan algorithm to identify frequent patterns in the transactions. The initial analysis without any restrictions returned single medications (e.g., Furosemida) and sequences of two medications (e.g., Varfarina,Furosemida), as expected. To discover more interesting results, we applied a filter to only show sequences with a minimum length of three medications and a minimum support of 5%.

To visualize the discovered patterns we used nodes to represent the medication and arrows (in different colors) to connect the sequences. For example, in figure 5.13 there are two patterns (represented by green and orange arrows) that begin with Atorvastatine, then in one pattern (orange) the next medication is Candesartan, while the other pattern (green) is followed by Propafenone, and both frequent patterns end in Warfarin.

Observing the frequent sequences from the dataset corresponding to age group 0-44 in figure 5.13 we can observe that old type anticoagulant Warfarin appears linked, as expected, to medications used to treat hypertension (Candesartan) and cholesterol (Atorvastatine) but also to less expected treatments like arrhythmia (Propafenone).

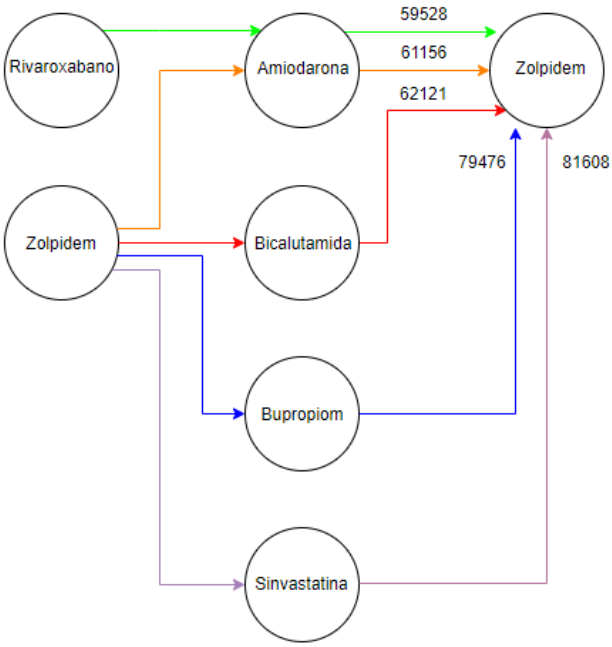
Looking at the frequent sequences from the dataset corresponding to age group 65-74 in figure 5.14 it is interesting to note that the NOAC named Rivaroxaban appears in an older age group linked with the treatment of cholesterol (Sinvastatine), and although not so expected, also with the treatment of insomnia (Zolpidem) representing more than 5,5% of the transactions in this group.

Our data-set corresponding to age group 45-54 did not return any frequent sequence with minimum support of 5% and with at least length 3. With regard to the other two data-sets (55-64 and 75+), they could not be processed in time to be evaluated and presented in this report due to the expansion of several records containing large number of items (≥ 10).



Sequences	Frequency	Percentage
Candesartan,Atorvastatina,Propafenona	12944	8,3%
Candesartan,Atorvastatina,Nebivolol	12944	8,3%
Candesartan,Atorvastatina,Candesartan	12712	8,1%
Atorvastatina,Candesartan,Varfarina	12368	7,9%
Atorvastatina,Propafenona,Varfarina	12288	7,9%

Figure 5.13: Top sequential patterns in age group 0-44.



Sequences	Frequency	Percentage
Zolpidem,Sinvastatina,Zolpidem	81608	7,7%
Zolpidem,Bupropiom,Zolpidem	79476	7,5%
Zolpidem,Bicalutamida,Zolpidem	62121	5,9%
Zolpidem,Amiodarona,Zolpidem	61156	5,8%
Rivaroxabano,Sinvastatina,Zolpidem	59528	5,6%

Figure 5.14: Top sequential patterns in age group 65-74.

6. Conclusions

In this chapter I present and discuss the final conclusions obtained from the evaluation and the contributions of this work, highlighting the implications drawn from its analysis and how some related problems can be approached in the future

6.1 Contributions

In this work I have defined methods to analyze treatment regimes from prescription databases covering entire populations that can be easily applied to other different contexts. To that end, I defined a novel methodology to transform a set of prescription records into a suitable database for data mining tasks, envisioning the discovery and exploration of relationships between prescribed medications.

Also, I have contributed with a thorough and detailed research on the state-of-the-art, studying the existing data mining techniques and evaluation methods, in particular those focused on mining prescriptions and other health record databases. Another important contribution of this work was the development of adequate visualization techniques to present the information the reader in a clear and concise way.

The results obtained from the analysis of a large database covering individuals prescribed with anticoagulants showed that regarding AR, the rules associated with the male patients have a greater lift when compared with the female ones. Also, it was interesting to note that not only the female top rules contain 33% more medications than the male's, but also that the NOAC Rivaroxaban already appears in the top rules. The comparison between the AR from Lisbon and Porto, the two biggest districts in terms of population, showed that Porto produces AR with much larger lift. With regards to the results for sequential patterns, it was possible to observe that in age group 0-44 Warfarin appears linked, as expected to medications used to treat hypertension and cholesterol, but also to less expected treatments like arrhythmia. Even more surprising was the connection found, in age group 65-74, between NOAC Rivaroxaban and medications used in insomnia, accounting for more than 5,5% of the transactions in this group.

6.2 Limitations and Recommendations for Future Work

Taking into account the focus of this study, as described in this dissertation, related future works on this subject may involve the definition of new types of transactions (e.g., Medical Doctor, Speciality or

Prescription) to provide different perspectives over the same data. I also recommend using the hierarchical classification of active substances to provide different levels of detail. Another recommendation I strongly suggest is the use of the Post Sequential Pattern Mining algorithm [Lu et al., 2008, Lu et al., 2010] to discover complex structural patterns hidden behind sequences, in order to represent relations between sequential patterns.

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